CHAPTER

1 Basic Math and Logarithm

NUMBER SYSTEM

(*i*) **Natural numbers:** The counting numbers 1, 2, 3, 4, ... are called Natural Numbers. The set of natural numbers is denoted by *N*.

Thus $N = \{1, 2, 3, 4, ...\}$.

(*ii*) **Whole numbers:** Natural numbers including zero are called whole numbers. The set of whole numbers is denoted by W.

Thus W = {0, 1, 2, ...}

(*iii*) **Integers:** The numbers ... – 3, – 2, – 1, 0, 1, 2, 3 .... are called integers and the set is denoted by I or Z. Thus I (or Z) = {.. – 3, – 2, – 1, 0, 1, 2, 3...}

Note: (*a*) Positive integers $I^+ = \{1, 2, 3 ....\} = N$

(*b*) Negative integers $I^- = \{....., –3, –2, –1\}$.

(*c*) Non-negative integers (whole numbers) = {0, 1, 2, ...}.

(*d*) Non-positive integers = {....., –3, –2, –1, 0}.

(*iv*) **Even integers:** Integers which are divisible by 2 are called even integers.

e.g. 0, ± 2, ± 4,.....

(*v*) **Odd integers:** Integers which are not divisible by 2 are called odd integers.

e.g. ± 1, ± 3, ± 5, ± 7.....

(*vi*) **Prime numbers:** Natural numbers which are divisible by 1 and itself only are called prime numbers.

e.g. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...

(*vii*) **Composite number:** Let 'a' be a natural number, '*a*' is said to be composite if, it has atleast three distinct factors.

e.g. 4, 6, 8, 9, 10, 12, 14, 15 .....

Note: (*a*) 1 is neither a prime number nor a composite number.

(*b*) Numbers which are not prime are composite numbers (except 1).

(*c*) '4' is the smallest composite number.

(*d*) '2' is the only even prime number.

(*viii*) **Co-prime numbers:** Two natural numbers (not necessarily prime) are called coprime, if their H.C.F (Highest common factor) is one.

e.g. (1, 2), (1, 3), (3, 4), (3, 10), (3, 8), (5, 6), (7, 8) (15, 16) etc.

These numbers are also called as **relatively prime** numbers.

Note:

(*a*) Two prime number(s) are always co-prime but converse need not be true.

(*b*) Consecutive natural numbers are always co-prime numbers.

(*ix*) **Twin prime numbers :** If the difference between two prime numbers is two, then the numbers are called twin prime numbers.

e.g. {3, 5}, {5, 7}, {11, 13}, {17, 19}, {29, 31}

Note: Number between twin prime numbers is divisible by 6 (except (3, 5)).

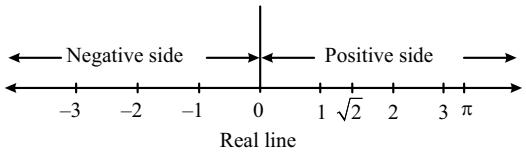
(*x*) **Rational numbers:** All the numbers that can be represented in the form p/q, where p and q are integers and $q \neq 0$, are called rational numbers and their set is denoted by Q. Thus $Q = \{p/q : p, q \in I \text{ and } q \neq 0\}$. It may be noted that every integer is a rational number since it can be written as p/q. It may be noted that all recurring decimals are rational numbers.

Note: Maximum number of different decimal digits in $\frac{p}{q}$ is equal to q, i.e. $\frac{11}{9}$ will have maximum of 9 different decimal digits.

(*xi*) **Irrational numbers:** The numbers which can not be expressed in p/q form where $p, q \in I$ and $q \neq 0$ i.e. the numbers which are not rational are called irrational numbers and their set is denoted by Q^c. (i.e. complementary set of Q) e.g. $\sqrt{2}$, $1 + \sqrt{3}$ etc. Irrational numbers can not be expressed as recurring decimals.

Note: $e \approx 2.71$ is called Napier's constant and $\pi \approx 3.14$ are irrational numbers.

(*xii*) **Real numbers:** Numbers which can be expressed on number line are called real numbers. The complete set of rational and irrational numbers is the set of real numbers and is denoted by R. Thus $R = Q \cup Q^C$.



All real numbers follow the order property i.e. if there are two distinct real numbers a and b then either $a < b$ or $a > b$.

Note:

(*a*) Integers are rational numbers, but converse need not be true.

(*b*) Negative of an irrational number is an irrational number.

(*c*) Sum of a rational number and an irrational number is always an irrational number

e.g. $2 + \sqrt{3}$

(*d*) The product of a non zero rational number and an irrational number will always be an irrational number.

(*e*) If $a \in Q$ and $b \notin Q$, then ab = rational number, only if $a = 0$.

(*f*) Sum, difference, product and quotient of two irrational numbers need not be a irrational number or we can say, result may be a rational number also.

(*xiii*) **Complex number:** A number of the form $a + ib$ is called a complex number, where $a, b \in R$ and $i = \sqrt{-1}$. Complex number is usually denoted by Z and the set of complex number is represented by C. Thus $C = \{a + ib : a, b \in R$ and $i = \sqrt{-1}\}$

Note: It may be noted that $N \subset W \subset I \subset Q \subset R \subset C$.

Train Your Brain

Example 1: The value of $1.\overline{285714} \div 1.\overline{714285} =$ _____.

(*a*) $\frac{3}{4}$ (*b*) $\frac{7}{8}$

(*c*) $\frac{7}{12}$ (*d*) $\frac{3}{7}$

Sol. (*a*)

$$1.\overline{285714}$$
$$= 1 + 0.\overline{285714}$$
$$= 1 + \frac{2}{7} = \frac{9}{7}$$
$$1.\overline{714285}$$
$$= 1 + \frac{5}{7} = \frac{12}{7}$$
$$\therefore 1.\overline{285714} \div 1.\overline{714285}$$
$$= \frac{9}{7} \div \frac{2}{7}$$
$$= \frac{9}{7} \times \frac{7}{12}$$
$$= \frac{3}{4}$$

Example 2: Prove that the difference $10^{25} - 7$ is divisible by 3.

Sol. Write the given difference in the form $10^{25} - 7 = (10^{25} - 1) - 6$. The number $10^{25} - 1 = \underbrace{\overline{99..9}}_{25 \text{ digits}}$ is divisible by 3 (and 9). Since the numbers $(10^{25} - 1)$ and 6 are divisible by 3, the number $10^{25} - 7$, being their difference, is also divisible by 3 without a remainder.

Concept Application

1. The product of $1.\overline{142857}$ and $0.\overline{63}$ = _____.

(*a*) $\frac{8}{11}$ (*b*) $\frac{7}{11}$

(*c*) $\frac{11}{7}$ (*d*) $\frac{8}{7}$

2. If $x = \sqrt{12} - \sqrt{9}, y = \sqrt{13} - \sqrt{10}$, and $z = \sqrt{11} - \sqrt{8}$, then which of the following is true?

(*a*) $z > x > y$

(*b*) $z > y > x$

(*c*) $y > x > z$

(*d*) $y > z > x$

SOME IMPORTANT IDENTITIES

1. $(a + b)^2 = a^2 + 2ab + b^2 = (a - b)^2 + 4ab$
2. $(a - b)^2 = a^2 - 2ab + b^2 = (a + b)^2 - 4ab$
3. $a^2 - b^2 = (a + b)(a - b)$
4. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
5. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
6. $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = (a + b)(a^2 + b^2 - ab)$
7. $a^3 - b^3 = (a - b)^3 + 3ab(a - b) = (a - b)(a^2 + b^2 + ab)$
8. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = a^2 + b^2 + c^2 + 2abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

9. $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$

10. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= \frac{1}{2}(a + b + c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$

If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

11. $a^4 - b^4 = (a + b)(a - b)(a^2 + b^2)$

12. $a^4 + a^2 + 1 = (a^2 + 1)^2 - a^2 = (1 + a + a^2)(1 - a + a^2)$

Train Your Brain

Example 3: Show that the expression, $(x^2 - yz)^3 + (y^2 - zx)^3 + (z^2 - xy)^3 - 3(x^2 - yz).(y^2 - zx).(z^2 - xy)$ is a perfect square and find its square root.e

Sol. $(x^2 - yz)^3 + (y^2 - zx)^3 + (z^2 - xy)^3 - 3(x^2 - yz)(y^2 - zx)(z^2 - xy) = a^3 + b^3 + c^3 - 3abc$

where $a = x^2 - yz, b = y^2 - zx, c = z^2 - xy$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= \frac{1}{2}(a + b + c)((a-b)^2 + (b-c)^2 + (c-a)^2)$

$= \frac{1}{2}(x^2 + y^2 + z^2 - xy - yz - zx)[(x^2 - yz - y^2 + zx)^2 + (y^2 - zx - z^2 + xy)^2 + (z^2 - xy - x^2 + yz)^2]$

$= \frac{1}{2}(x^2 + y^2 + z^2 - xy - yz - zx)[\{x^2 - y^2 + z(x-y)\}^2 + \{y^2 - z^2 + x(y-z)\}^2 + \{z^2 - x^2 + y(z-x)\}^2]$

$= \frac{1}{2}(x^2 + y^2 + z^2 - xy - yz - zx)(x + y + z)^2 [(x-y)^2 + (y-z)^2 + (z-x)^2]$

$= (x + y + z)^2 (x^2 + y^2 + z^2 - xy - yz - zx)^2$

$= (x^3 + y^3 + z^3 - 3xyz)^2$

(which is a perfect square) its square roots are

$\pm(x^3 + y^3 + z^3 - 3xyz)$

Example 4. If $x^2 - 4x + 1 = 0$, then what is the value of $x^3 + \frac{1}{x^3}$?

Sol. $x^2 - 4x + 1 = 0$

$\Rightarrow x = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 1}}{2 \times 1} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$

$\therefore x^3 + \frac{1}{x^3} = \left(2 + \sqrt{3}\right)^3 + \frac{1}{\left(2 + \sqrt{3}\right)^3} = \left(2 + \sqrt{3}\right)^3 + \left[\frac{\left(2 - \sqrt{3}\right) \times 1}{\left(2 + \sqrt{3}\right)\left(2 - \sqrt{3}\right)}\right]^3 = \left(2 + \sqrt{3}\right)^3 + \left(2 - \sqrt{3}\right)^3$

$= 2^3 + \left(\sqrt{3}\right)^3 + 3 \times 2 \times \sqrt{3}\left(2 + \sqrt{3}\right) + 2^3 - \left(\sqrt{3}\right)^3 - 3 \times 2 \times \sqrt{3}\left(2 - \sqrt{3}\right)$

$= 8 + 18 + 8 + 18 = -50$. Similarly for

$x = 2 - \sqrt{3}, x^3 + \frac{1}{x^3} = 52$

Example 5. If $x = \frac{1}{x} = a$, then what is the value of $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$?

(a) $a^3 + a^2$ (b) $a^3 + a^2 - 5a$

(c) $a^3 + a^2 - 3a - 2$ (d) $a^3 + a^2 - 4a - 2$

Sol. (c)

Given, $x + \frac{1}{x} = a$

Now, $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2} = \left(x^3 + \frac{1}{x^3}\right) + \left(x^2 + \frac{1}{x^2}\right)$

$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right)^2 - 2$

$= a^3 - 3a + a^2 - 2 = a^3 + a^2 - 3a - 2$.

Concept Application

3. If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then what is $(x+y+z)^3$ equal to?

(a) 1 (b) 3

(c) $3xy$ (d) $27xyz$

4. If $a + b + c = 0$, then what is the value of

$$\frac{a^2 + b^2 + c^2}{(a-b)^2 + (b-c)^2 + (c-a)^2}$$

(a) 1 (b) 3 (c) $\frac{1}{3}$ (d) 0

5. If $x + \frac{1}{x} = p$ then $x^6 + \frac{1}{x^6}$ equals to :

(a) $p^6 + 6p$ (b) $p^6 - 6p$

(c) $p^6 + 6p^4 + 9p^2 + 2$ (d) $p^6 - 6p^4 + 9p^2 - 2$

6. If $x + \frac{1}{x} = 4$, then find values of

(i) $x^2 + \frac{1}{x^2}$ (ii) $x^3 + \frac{1}{x^3}$

(iii) $x^4 + \frac{1}{x^4}$

7. Prove that $(1 + x)(1 + x^2)(1 + x^4)(1 + x^8)(1 + x^{16}) = \frac{(1 - x^{32})}{(1 - x)}$

8. If x, y, z are all different real numbers, then prove that

$$\frac{1}{(x-y)^2}+\frac{1}{(y-z)^2}+\frac{1}{(z-x)^2}=\left(\frac{1}{x-y}+\frac{1}{y-z}+\frac{1}{z-x}\right)^2$$

9. If $\frac{a}{b}+\frac{b}{a}=-1$, then find value of $a^3 - b^3$.

10. If $a - b = -8$, $ab = -12$ then $a^3 - b^3$ will be

11. The product $(x + y)(x - y)(x^2 + xy + y^2)(x^2 - xy + y^2)$ simplifies to

12. Find the real values of p, q, r satisfying $(2p - 3)^8 + (1 - q)^6 + (4 - 3r)^4 = 0$.

INDICES

If 'a' is any non zero real or imaginary number and 'm' is the positive integer, then $a^m = a \cdot a \cdot a \ldots a$ (m times). Here a is called the base and m is called the index, power or exponent.

Law of indices:

1. $a^0 = 1, (a \neq 0)$
2. $a^{-m} = \frac{1}{a^m}, (a \neq 0)$
3. $a^{m+n} = a^m \cdot a^n$, where m and n are rational numbers
4. $a^{m-n} = \frac{a^m}{a^n}$, where m and n are rational numbers, $a \neq 0$
5. $(a^m)^n = a^{mn}$
6. $a^{p/q} = \sqrt[q]{a^p}$
7. $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a} = \sqrt[n]{\sqrt[m]{a}}$, where $m, n \in N$ and $(m, n \geq 2)$ and a is positive rational number
8. $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$, $a, b \in R$ and atleast one of a and b should be positive

SURDS

If a is a positive rational number, which is not the nth power (n is any natural number) of any rational number, then the irrational number $\pm\sqrt[n]{a}$ are called simple surds or monomial surds.

Every surd is an irrational number (but every irrational number is not a surd). So, the representation of monomial surd on a number line is same as that of irrational numbers.

Examples:

1. $\sqrt{3}$ is a surd and $\sqrt{3}$ is an irrational number.

2. $\sqrt[3]{5}$ is a surd and $\sqrt[3]{5}$ is an irrational number.

3. π is an irrational number, but it is not a surd.

4. $\sqrt[3]{3+\sqrt{2}}$ is an irrational number. It is not a surd, because $3+\sqrt{2}$ is not a rational number.

Train Your Brain

Example 6: Simplify $\left[\sqrt[3]{\sqrt[6]{a^9}}\right]^4 \left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$

Sol. $a^{9(1/6)(1/3)4} \cdot a^{9(1/3)(1/6)4} = a^2 \cdot a^2 = a^4$.

Example 7: Arrange the following in ascending or descending order of magnitude:

$\sqrt[4]{6}, \sqrt[3]{7}, \sqrt{5}$

Sol. $\sqrt[4]{6} = 6^{1/4}, \sqrt[3]{7} = 7^{1/3}, \sqrt{5} = 5^{1/2}$

LCM of the denominators of the exponents of these three terms, 4, 3 and 2 is12.

Now express the exponent of each term, as a fraction in which then denominator is 12.

$$6^{\frac{1}{4}} = 6^{\frac{3}{12}} = \left(6^3\right)^{\frac{1}{12}} = \sqrt[12]{216}$$

$$7^{\frac{1}{3}} = 7^{\frac{4}{12}} = \left(7^4\right)^{\frac{1}{12}} = \sqrt[12]{2401}$$

$$5^{\frac{1}{2}} = 5^{\frac{6}{12}} = \left(5^6\right)^{\frac{1}{12}} = \sqrt[12]{15625}$$

Now, $\sqrt[4]{6} = \sqrt[12]{216}, \sqrt[3]{7} = \sqrt[12]{2401}, \sqrt{5} = \sqrt[12]{15625}$

Hence, their ascending order is

$\sqrt[12]{216}, \sqrt[12]{2401}, \sqrt[12]{15625}$, *i.e.*, $\sqrt[4]{6}, \sqrt[3]{7}, \sqrt{5}$

$\therefore$ The descending order of magnitude of the given radical is $\sqrt{5}, \sqrt[3]{7}, \sqrt[4]{6}$.

Example 8: Find the square root of $10+\sqrt{24}+\sqrt{60}+\sqrt{40}$

Sol.

$$= \sqrt{10+\sqrt{24}+\sqrt{60}+\sqrt{40}}$$

$$= \sqrt{10+2\sqrt{6}+2\sqrt{15}+2\sqrt{10}}$$

$$= \sqrt{(2+3+5)+2\sqrt{2(3)}+2\sqrt{3(5)}+2\sqrt{2(5)}}$$

$$= \sqrt{\left(\sqrt{2}+\sqrt{3}+\sqrt{5}\right)^2}$$

$$= \sqrt{2}+\sqrt{3}+\sqrt{5}.$$

Concept Application

13. If the surds $\sqrt[4]{4}, \sqrt[6]{5}, \sqrt[8]{6}$ *and* $\sqrt[12]{8}$ are arranged in ascending order from left to right, then the third surd from the left is

(*a*) $\sqrt[12]{8}$ (*b*) $\sqrt[4]{4}$

(*c*) $\sqrt[8]{6}$ (*d*) $\sqrt[6]{5}$

14. $\sqrt[6]{15-2\sqrt{56}}\sqrt[3]{\sqrt{7}+2\sqrt{2}}$ = _____.

(*a*) 0 (*b*) $\sqrt{2}$

(*c*) 1 (*d*) $\sqrt[6]{2}$

15. If $x = 2^{1/3} - 2$, then $x^3 + 6x^2 + 12x$ = _____.

(*a*) 6 (*b*) –6

(*c*) 8 (*d*) –8

16. If $a = 4^{\frac{-3}{2}}$, $b = (125)^{\frac{-2}{3}}$, $c = 81^{\frac{-3}{4}}$ then $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ =

17. If $p = 8^{\frac{2}{3}}$, $q = 9^{\frac{3}{2}}$, $r = 16^{\frac{5}{4}}$, $s = \frac{1}{6^{-2}}$ then $p + q + r + s$ =

18. If $u = \left(\frac{1}{27}\right)^{\frac{-4}{3}}$, $v = \left(\frac{1}{216}\right)^{\frac{-2}{3}}$, $w = (8)^{5/3} \cdot (4)^{-5/2}$, $t = (9)^{3/2} \cdot (81)^{-3/4}$ then $\frac{vwt}{\sqrt{u}}$ =

19. Which of the following number is greater than 1

(*a*) $\left(\frac{1}{216}\right)^{-2/3} \cdot \left(\frac{1}{6^{-2}}\right)$ (*b*) $(125)^{-2/3} \cdot (625)^{1/2}$

(*c*) $\log_{1/2} 16$ (*d*) $\log_{16} 32$

20. Simplify $a\left(\frac{\sqrt{a}+\sqrt{b}}{2b\sqrt{a}}\right)^{-1} + b\left(\frac{\sqrt{a}+\sqrt{b}}{2a\sqrt{b}}\right)^{-1}$

RATIO

(*i*) If A and B be two quantities of the same kind, then their ratio is $A : B$; which may be denoted by the fraction $\frac{A}{B}$ (This may be an integer or fraction)

(*ii*) A ratio may represented in a number of ways e.g. $\frac{a}{b} = \frac{ma}{mb} = \frac{na}{nb}$ =..... where m, n,..... are non-zero numbers.

(*iii*) To compare two or more ratio, reduce them to common denominator.

(*iv*) Ratio between two ratios may be represented as the ratio of two integers

e.g. $\frac{a}{b} : \frac{c}{d} : \frac{a/b}{c/d} = \frac{ad}{bc}$ or $ad : bc$

(*v*) Ratios are compounded by multiplying them together i.e. $\frac{a}{b} \cdot \frac{c}{d} \cdot \frac{e}{f} ..... = \frac{ace}{bdf} .....$

(*vi*) If $a : b$ is any ratio then its duplicate ratio is $a^2 : b^2$; triplicate ratio is $a^3 : b^3$..... etc.

(*vii*) If $a : b$ is any ratio, then its sub-duplicate ratio is $a^{1/2} : b^{1/2}$; sub-triplicate ratio is $a^{1/3} : b^{1/3}$ etc.

PROPORTION

When two ratios are equal, then the four quantities compositing them are said to be proportional. If $\frac{a}{b} = \frac{c}{d}$, then it is written as $a : b = c : d$ or $a : b :: c : d$

(*i*) 'a' and 'd' are known as extremes and 'b and c' are known as means.

(*ii*) An important property of proportion Product of extremes = product of means.

(*iii*) If $a : b = c : d$, then $b : a = d : c$ (Invertando)

(*iv*) If $a : b = c : d$, then $a : c = b : d$ (Alternando)

(*v*) If $a : b = c : d$, then $\frac{a+b}{b} = \frac{c+d}{d}$ (Componendo)

(*vi*) If $a : b = c : d$, then $\frac{a-b}{b} = \frac{c-d}{d}$ (Dividendo)

(*vii*) If $a : b = c : d$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ (Componendo and dividendo)

(*viii*) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = ...$, then each $\frac{a+c+e+.....}{b+d+f+.....}$ $= \frac{\text{Sum of the numerators}}{\text{Sum of the denominators}}$

(*ix*) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = ...$, then each = $\frac{xa+yc+ze+.....}{xb+yd+zf+.....}$

(*x*) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = ...$, then each = $\left(\frac{xa^n+yc^n+ze^n}{xb^n+yd^n+zf^n}\right)^{1/n}$

Train Your Brain

Example 9: If $\frac{x+y}{2} = \frac{y+z}{3} = \frac{z+x}{4}$, then find $x : y : z$.

Sol. Each = $\frac{\text{Sum of the numerators}}{\text{Sum of the denominators}}$

$= \frac{2(x+y+z)}{9} = \frac{x+y+z}{9/2}$ and therefore each

$= \frac{(x+y+z)-(y+z)}{\frac{9}{2}-3} = \frac{(x+y+z)-(x+z)}{\frac{9}{2}-4}$

$= \frac{(x+y+z)-(x+y)}{\frac{9}{2}-2}$

$= \frac{x}{3/2} = \frac{y}{1/2} = \frac{z}{5/2} \Rightarrow x : y : z = 3 : 1 : 5$

Example 10: If $\frac{a+3b+2c+6d}{-a-3b+2c+6d} = \frac{3a+b+6c+2d}{-3a-b+6c+2d}$, then the correct statement is

(*a*) $ad = bc$ (*b*) $ac = bd$

(*c*) $c = \frac{ab}{d}$ (*d*) $a + d = b + c$

Sol. (a) Apply C and D the the given equation

$$\Rightarrow \frac{4c+12d}{2a+6d}=\frac{4d+12c}{2b+6a} \quad \Rightarrow \frac{c+3d}{a+3b}=\frac{d+3c}{b+3a}$$

$\Rightarrow bc + 3ac + 3bd + 9ad = ad + 3ac + 3bd + 9bc$

$\Rightarrow 8ad = 8bc$

$\Rightarrow ad = bc$



Concept Application

21. Solve the equations $\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}=\frac{2x^2-1}{2}$

22. Solve: $\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}=3$

23. If $(4x - 3y) : (2x + 5y) = 12 : 19$ then $x : y$ is .....

(a) 2 : 3 (b) 1 : 2

(c) 3 : 2 (d) 2 : 1

INTERVALS

Intervals are basically subsets of R and are commonly used in solving inequalities or in finding domains. If there are two numbers $a, b \in R$ such that $a < b$, we can define four types of intervals as follows :

Open Interval (a, b)

$\{x : a < x < b\}$ i.e. extreme points are not includes

Closed Interval $[a, b]$

$\{x : a \le x \le b\}$ i.e. extreme points are includes

It can possible when a and b are finite

Semi-Open Interval $(a, b]$

$\{x : a < x \le b\}$ i.e. a is not include and b is include

Semi-Closed Interval $[a, b)$

$\{x : a \le x < b\}$ i.e. a is include and b is not include

Note:

1. The infinite intervals are defined as follows :
   (i) $(a, \infty) = \{x : x > a\}$
   (ii) $[a, \infty) = \{x : x \ge a\}$
   (iii) $(-\infty, b) = \{x : x < b\}$
   (iv) $(\infty, b] = \{x : x \le b\}$
   (v) $(-\infty, \infty) = \{x : x \in R\}$
2. $x \in \{1, 2\}$ denotes some particular values of x, i.e. $x = 1, 2$
3. If there is no value of x, then we say $x \in \phi$ (null set)

GENERAL METHOD TO SOLVE INEQUALITIES

Method of Intervals (Wavy Curve Method)

Let $g(x) = \left(\frac{(x-b_1)^{k_1}(x-b_2)^{k_2}---(x-b_n)^{k_n}}{(x-a_1)^{r_1}(x-a_2)^{r_2}---(x-a_n)^{r_n}}\right)$ ...(i)

Where $k_1, k_2 ..... k_n$ and $r_1, r_2 ..... r_n \in N$ and $b_1, b_2..... b_n$ and $a_1, a_2 ... a_n$ are real numbers.

Then to solve the inequality following steps are taken.

Steps: Points where numerator becomes zero are called zeros or roots of the function and where denominator becomes zero are called poles of the function.

(i) First we find the zeros and poles of the function.

(ii) Then we mark all the zeros and poles on the real line and put a vertical bar to divide the real line in many intervals.

(iii) Determine sign of the function in any of the interval and then alternates the sign in the neghbouring interval if the poles or zeros dividing the two interval has appeared odd number of times otherwise retain the sign.

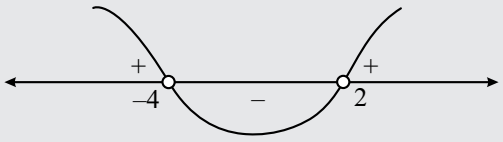
(iv) Thus we consider all the intervals. The solution of the $g(x) > 0$ is the union of the intervals in which we have put the plus sign and the solution of $g(x) < 0$ is the union of all intervals in which we have put the minus sign.



Train Your Brain

Example 11: Solution $\frac{(x-1)^2(x+4)}{(2-x)} < 0$ is

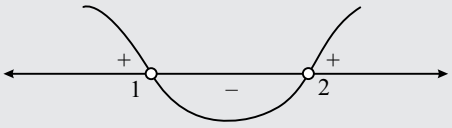
Sol. $\frac{(x-1)^2(x+4)}{(2-x)} < 0 \Rightarrow \frac{(x+4)}{(x-2)} > 0$



$\Rightarrow (-\infty, -4) \cup (2, \infty)$.

Example 12: The solution of $\frac{\sqrt[3]{(x+4)^4}(x-1)^3}{(x-2)} > 0$ is

Sol. $\frac{(x+4)^{4/3}(x-1)^3}{(x-2)} > 0 \Rightarrow \frac{x-1}{x-2} > 0$



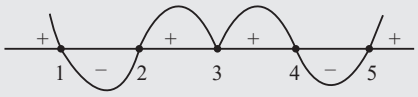
$\Rightarrow (-\infty, 1) \cup (2, \infty)$ Excluding – 4.

Example 13. Find the range of x, so that following expressions are defined.

(a) $(x-1)(x-2)(x-3)^2(x-4)^5(x-5) > 0$

(b) $\dfrac{(x-1(x-2)}{(x-3)} \geq 0$

Sol. (a) $\Rightarrow x \in (-\infty,1) \cup (2,3) \cup (3,4) \cup (5,\infty)$



(b) $\Rightarrow x \in [1,2] \cup (3,\infty)$



Concept Application

24. Solve the inequality $\dfrac{x^3-4x^2-12x}{x+3} \geq 0$

25. Solve the inequality $\dfrac{(x+1)^4(x-2)(x-3)^3(x-4)^2}{x^3-36x} \geq 0$

26. Find the number of integer values of variable x satisfying the following pair of inequalities.

$\dfrac{(x-1)(x+4)}{x-3} < 0 \;\&\; x^2+6x-27 \leq 0$

27. The solution of the inequality $2x-1 \leq x^2+3 \leq x-1$ is

(a) $x \in R$ (b) $(-2, 2]$

(c) $(-2, 2)$ (d) $x \in \phi$

LOGARITHM FUNCTION

Introduction

The kinetics of some chemical reactions (which rely on the concentration of one or more of the reactants), radioactive decay (used in carbon dating), are examples of exponential decay.

A nautilus's shell develops in such a way that each new chamber is roughly a replica of the previous one but is enlarged by a fixed amount. This results in the creation of the elegant shape known as a logarithmic spiral. The pace of growth is related to the existing size of many objects in nature. So, it is extremely important to comprehend the mathematics that explains the behaviour of entities like birth rates if we are to forecast how they will behave in the future. Logarithmic scales (the Richter scale in the case of an earthquake) are useful in offering a means to handle the numerical values of these measures.

Definition

The logarithm of the number N to the base 'a' is the exponent indicating the power to which the base 'a' must be raised to obtain the number N.

This number is designated as $\log_a N$.

Hence $\log_a N = x \Leftrightarrow a^x = N,\ a > 0,\ a \neq 1$ and $N > 0$

If $a = 10$, then we write $\log b$ rather than $\log_{10} b$

$a = e$, we write $\ln b$ rather than $\log_e b$

The existence and uniqueness of the number $\log_a N$ follows from the properties of an exponential functions.

Domain

The existence and uniqueness of the number $\log_a N$ can be determined with the help of set of conditions, $a > 0$ & $a \neq 1$ & $N > 0$.

FUNDAMENTAL IDENTITY

(*i*) $\log_a 1 = 0$ $(a > 0,\ a \neq 1)$

(*ii*) $\log_a a = 1$ $(a > 0,\ a \neq 1)$

(*iii*) $\log_{1/a} a = -1$ $(a > 0,\ a \neq 1)$

Remember

$\log_{10} 2 \approx 0.3010$; $\log_{10} 3 \approx 0.4771$

$\ln 2 \approx 0.693$; $\ln 10 \approx 2.303$

FUNDAMENTAL LOGARITHMIC IDENTITY

$a^{\log_a N} = N,\ a > 0,\ a \neq 1 \;\&\; N > 0$

Proof :

$\log_a N = x$ ... (i)

$N = (a)^x$ ... (ii)

by equation (i) & (ii)

$N = (a)^{\log_a N}$

PRINCIPAL PROPERTIES

Let M & N are arbitrary positive numbers, $a > 0$, $a \neq 1$, and x, y are any real numbers, then:

(*i*) $\log_a (M \cdot N) = \log_a M + \log_a N$;

Proof:

Let $\log_a M = x$ and $\log_a N = y$

$\Rightarrow M = a^x$ and $N = a^y$

Now, $MN = a^x a^y = a^{x+y} \Rightarrow \log_a MN = x + y$

In general

$\log_a (x_1 x_2 \ldots x_n) = \log_a x_1 + \log_a x_2 + \ldots + \log_a x_n$

(*ii*) $\log_a (M/N) = \log_a M - \log_a N$

Proof:

Let $\log_a M = x$ and $\log_a N = y$

$\Rightarrow M = a^x$ and $N = a^y$

Now, $M/N = a^x/a^y = a^{x-y} \Rightarrow \log_a M/N = x - y$

(*iii*) $\log_a M^\alpha = \alpha \cdot \log_a M$

BASE CHANGING THEOREM

It states that ratio of logarithm of two numbers is independent of their common base

Symbolically

$$\frac{\log_a M}{\log_a b} = \log_b M \ (a > 0, M > 0, b > 0)$$

Proof:

Let $\log_b M = x$

$\Rightarrow M = b^x$

$\Rightarrow \log_a M = \log_a b^x$

$\Rightarrow \log_a M = x \cdot \log_a b$

$\Rightarrow \dfrac{\log_a M}{\log_a b} = x = \log_b M$

Important Results

(*i*) **Base power formula:** $\log_{a^k} M = \dfrac{1}{k} \log_a M$

Proof:

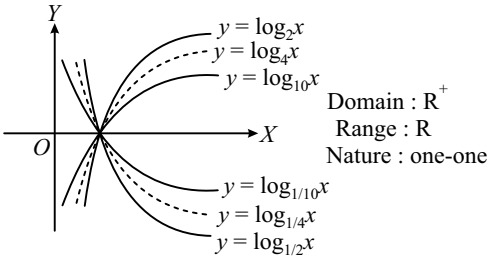
$$\log_{a^k}(M) = \frac{\log_a M}{\log_a a^k} = \frac{\log_a M}{k \log_a a} = \frac{1}{k} \log_a M$$

(*ii*) $a^{\log_b c} = c^{\log_b a}$

Proof: $a^{\log_b c} = a^{\log_a c . \log_b a} = (a^{\log_a c})^{\log_b a} = (c)^{\log_b a}$

GRAPH OF LOGARITHMIC FUNCTIONS

If $a > 0, a \neq 1$, then the function $y = \log_a x, x \in R^+$ (set of positive real numbers) is called the logarithmic Function with base a.

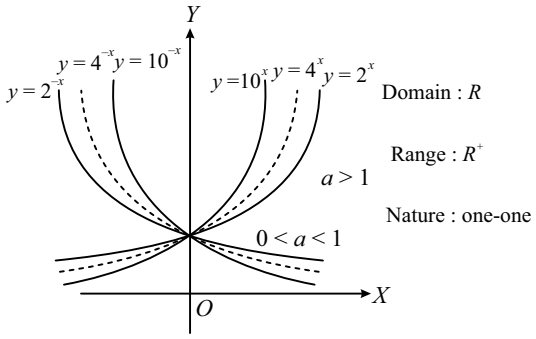


Note: (*i*) If the number and the base are on the same side of the unity, then the logarithm is positive.

(*ii*) If the number and the base are on the opposite sides of unity, then the logarithm is negative.

EXPONENTIAL FUNCTION

If $a > 0, a \neq 1$ then the function defined by $f(x) = a^x, x \in R$ is called an Exponential Function with base a.



LOGARITHMIC EQUATION

The equality $\log_a x = \log_a y$ is possible if and only if $x = y$

i.e. $\log_a x = \log_a y \Leftrightarrow x = y$

Always check validity of given equation, ($x > 0, y > 0, a > 0, a \neq 1$)

LOGARITHMIC INEQUALITY

Let 'a' is a real number such that

(*i*) If $a > 1$, then $\log_a x > \log_a y \Rightarrow x > y$

(*ii*) If $a > 1$, then $\log_a x < \alpha \Rightarrow 0 < x < a^\alpha$

(*iii*) If $a > 1$, then $\log_a x > \alpha \Rightarrow x > a^\alpha$

(*iv*) If $0 < a < 1$, then $\log_a x > \log_a y \Rightarrow 0 < x < y$

(*v*) If $0 < a < 1$, then $\log_a x < \alpha \Rightarrow x > a^\alpha$

Form - I : $f(x) > 0, g(x) > 0, g(x) \neq 1$

Form	**Collection of system**
(*a*) $\log_{g(x)} f(x) \geq 0 \Leftrightarrow$	$\begin{cases} f(x) \geq 1, & g(x) > 1 \\ 0 < f(x) \leq 1, & 0 < g(x) < 1 \end{cases}$
(*b*) $\log_{g(x)} f(x) \leq 0 \Leftrightarrow$	$\begin{cases} f(x) \geq 1 & , \ 0 < g(x) < 1 \\ 0 < f(x) \leq 1 & , \ g(x) > 1 \end{cases}$
(*c*) $\log_{g(x)} f(x) \geq a \Leftrightarrow$	$\begin{cases} f(x) \geq g(x)^a & , \ g(x) > 1 \\ 0 < f(x) \leq g(x)^a & , \ 0 < g(x) < 1 \end{cases}$
(*d*) $\log_{g(x)} f(x) \leq a \Leftrightarrow$	$\begin{cases} 0 < f(x) \leq g(x)^a & , \ g(x) > 1 \\ f(x) \geq g(x)^a & , \ 0 < g(x) < 1 \end{cases}$

From - II : $f(x) > 0, g(x) > 0, \phi(x) > 0, \phi(x) \neq 1$

Form	**Collection of system**
(*a*) $\log_{\phi(x)} f(x) \geq \log_{\phi(x)} g(x) \Leftrightarrow$	$\begin{cases} f(x) \geq g(x), \ \phi(x) > 1, \\ 0 < f(x) \leq g(x); 0 < \phi(x) < 1 \end{cases}$
(*b*) $\log_{\phi(x)} f(x) \leq \log_{\phi(x)} g(x) \Leftrightarrow$	$\begin{cases} 0 < f(x) \leq g(x), \phi(x) > 1, \\ f(x) \geq g(x) > 0, \ 0 < \phi(x) < 1 \end{cases}$

COMMON AND NATURAL LOGARITHM

$\log_{10}N$ is referred as a common logarithm and $\log_e N$ is called as natural logarithm of N to the base Napierian and is popularly written as $\ell n\, N$. Note that e is an irrational quantity lying between 2.7 to 2.8 **Note that** $e^{\ell n\, x} = x$.



Train Your Brain

Example 14: How many solutions are there for equation

$\log_4 (x-1) = \log_2 (x-3)$?

Sol. $\log_4 (x-1) = \log_2 (x-3)$

$\Rightarrow \log_{2^2} (x-1) = \log_2 (x-3)$

$\Rightarrow \frac{1}{2} \log_2 (x-1) = \log_2 (x-3)$

$\Rightarrow \log_2 (x-1)^{1/2} = \log_2 (x-3)$

$\Rightarrow (x-1)^{1/2} = (x-3)$

$\Rightarrow x - 1 = x^2 - 6x + 9$

$\Rightarrow (x-2)(x-5) = 0$

$\Rightarrow x = 2, 5$

But $x - 1 > 0$ and $x - 3 > 0$

$x > 1$ and $x > 3$

So only one solution $x = 5$

Example 15: Solve the logarithmic inequality $\log_{1/5} \frac{4x+6}{x} \geq 0$.

Sol. Since $\log_{1/5} 1 = 0$, the given inequality can be written as.

$$\log_{1/5} \frac{4x+6}{x} \geq \log_{1/5} 1$$

When the domain of the function is taken into account the inequality is equivalent to the system of inequalities.

$$\begin{cases} \frac{4x+6}{x} > 0, \\ \frac{4x+6}{x} \leq 1 \end{cases}$$

Solving the inequalities by using method of intervals

$$x \in \left[-2, \frac{-3}{2}\right)$$

Example 16: For $x \geq 0$, what is the smallest possible value of the expression $\log(x^3 - 4x^2 + x + 26) - \log(x + 2)$?

Sol. $\log \frac{(x^3 - 4x^2 + x + 26)}{(x+2)}$

$= \log \frac{(x^2 - 6x^2 + 13)(x+2)}{(x+2)}$

$= \log (x^2 - 6x + 13)$ $[\because x \neq -2]$

$= \log\{(x-3)^2 + 4\}$

$\therefore$ Minimum value is log 4 when $x = 3$

Example 17: Given $\log_2 a = s$, $\log_4 b = s^2$ and $\log_{c^2}(8) = \frac{2}{s^3+1}$.

Write $\log_2 \frac{a^2 b^5}{c^4}$ as a function of 's' ($a, b, c > 0, c \neq 1$).

Sol. Given $\log_2 a = s$ ...(i)

$\log_2 b = 2s^2$ ...(ii)

$\log_8 c^2 = \frac{s^3+1}{2}$ ...(iii)

$\Rightarrow \frac{2\log c}{3\log 2} = \frac{s^3+1}{2} \Rightarrow 4\log_2 c = 3(s^3+1)$ ...(iv)

to find $2\log_2 a + 5\log_2 b - 4\log_2 c$

$\Rightarrow 2s + 10s^2 - 3(s^3+1)$

Example 18. If $\frac{1}{4}\log_4 M + 4\log_4 N = 1 + \log_{.008} 5$ then the value of $MN^{16} = k.2^{1/3}$, where k is equal to

(a) 8 (b) 32

(c) 36 (d) 40

Sol. (b) $\frac{1}{8}\log_2 M + 2\log_2 N = 1 + \frac{\log_2 5}{\log_2 (.008)}$

$\Rightarrow \log_2 M^{1/8} + \log_2 N^2 = 1 + \frac{\log_2 10 - 1}{3 - 3\log_2 10}$

$\Rightarrow \log_2 \left(MN^{16}\right)^{1/8} = 1 - \frac{1}{3} = \frac{2}{3}$

$\Rightarrow \left(MN^{16}\right)^{1/8} = 2^{2/3}$

$\Rightarrow MN^{16} = 2^{16/3} = 32(2^{1/3})$



Concept Application

28. Solve the inequality $\log_{1/3} (5x - 1) > 0$.

29. Suppose that a and b are positive real numbers such that $\log_{27} a + \log_9 b = \frac{7}{2}$ and $\log_{27} b + \log_9 a = \frac{2}{3}$. Find the value of the ab.

30. If $m_1 = \log_8 16$, $m_2 = \log_{81} 27$, $m_3 = \log_{1/3} 1/9$, $m_4 = \log_{1/3} 9\sqrt{3}$ then $m_1 \cdot m_2 \cdot m_3 \cdot m_4$

31. If $q = \log_{2\sqrt{3}} 1728$, $q = \log_2 (\cos 45°)$, $r = \log_2 (\log_2 4)$, $s = \log_3 (\tan 30°)$, $t = \log_{625} 125$ then $\frac{prt}{qs} =$

32. If $\log_7(\log_3(\log_2 x)) = 0$, then find $\log_{0.125} x$.

33. Solve for x:

(i) $\log_3 x > 0$ (ii) $\log_5 x \geq 0$

(iii) $\log_6 x < 0$ (iv) $\log_2 x \leq 0$

(v) $\log_{1/7} x > 0$ (vi) $\log_{1/8} x \geq 0$

(vii) $\log_{1/9} x < 0$ ($viii$) $\log_{1/e} x \leq 0$

(ix) $\log_2(x-1) > 1$ (x) $\log_{1/2}(x-2) \leq 1$

34. Solve for x:

(i) $\log_4(2x-3) < 2$ (ii) $\log_{1/2}(3x-2) \geq 3$

(iii) $\log_{16}(\log_4(x)) > 1$ (iv) $\log_{1/2}(\log_{1/4}(x)) < 1$

35. $\log_2(a^2-5) = 2$

36. $\log_{1/3}(a^2-1) = -1$

37. Prove that : $2^{\sqrt{\log_2 3}} = 3^{\sqrt{\log_3 2}}$

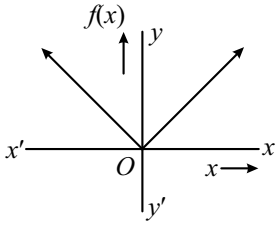
38. If $\log_{12}27 = a$ find the value of $\log_6 16$ in term of a.

ABSOLUTE VALUE FUNCTION / MODULUS FUNCTION

This is also known as absolute value function and denoted by

$$f(x) = |x| \text{ i.e. } f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Domain of this function is set of all real numbers because $f(x)$ exists for all $x \in R$ but $|x| \geq 0$ so range is all non-negative real numbers.



Domain = R; Range = $[0, \infty)$ or $R^+ \cup \{0\}$

Properties of modulus : For any $a, b \in R$

(i) $|a| \geq 0$

(ii) $|a| = |-a|$

(a) $|a|^n = |a^n|$

(b) $|a^n| = a^n$, where n is even and $n \in z$

(iii) $|a| \geq a, |a| \geq -a$

(iv) $|ab| = |a||b|$

(v) $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

Note: $|f(x)| + |g(x)| = |f(x) + g(x)|$

$\Rightarrow f(x) \cdot g(x) \geq 0$

Train Your Brain

Example 19 The absolute value of sum of real solutions of $\log_2|x^2+5x+4| = \log_2 3 + \log_2|x+1|$ is

(a) 8 (b) 6 (c) 7 (d) 5

Sol. $\log_2 \frac{|(x+1)(x+4)|}{|x+1|} = \log_2 3$

$|x+4| = 3$

$x+4 = -3, +3$

$x = -7, -1$ (rejected); $\Rightarrow x = -7$

Example 20: Solve the following linear equation

(i) $x|x| = 4$

(ii) $|x-3| + 2|x+1| = 4$

Sol. (i) $x|x| = 4$

If $x > 0$

$\therefore x^2 = 4 \Rightarrow x = \pm 2 \quad \therefore x = 2 \ (\because x \geq 0)$

If $x < 0 \Rightarrow -x^2 = 4 \Rightarrow x^2 = -4$ which is not possible

(ii) $|x-3| + 2|x+1| = 4$

Case-I If $x \leq -1$

$\therefore -(x-3) - 2(x+1) = 4$

$\Rightarrow -x + 3 - 2x - 2 = 4 \Rightarrow -3x + 1 = 4$

$\Rightarrow -3x = 3 \Rightarrow x = -1$

Case-II If $-1 < x \leq 3$

$\therefore -(x-3) + 2(x+1) = 4$

$\Rightarrow -x + 3 + 2x + 2 = 4$

$\Rightarrow x = -1$ which is not possible

Case-III If $x > 3$

$x - 3 + 2(x+1) = 4$

$3x - 1 = 4 \Rightarrow x = 5/3$ which is not possible

$\therefore x = -1$

Example 21: Number of real solutions of $|x-1| = |x-2| + |x-3|$ is

(a) 0 (b) 1

(c) 2 (d) more than 2

Sol.

Case-I: $x \leq 1, 1 - x = 2 - x + 3 - x$

$x = 4$ (rejected)

Case-II: $1 < x \leq 2, x - 1 = 2 - x + 3 - x, x = 2$

Case-III: $2 < x < 3, x - 1 = x - 2 + 3 - x, x = 2$

Case-IV: $x \geq 3, x - 1 = x - 2 + x - 3$

$x = 4 \quad \Rightarrow x = 2, 4$

Concept Application

39. Let $x_1, x_2, x_3, x_4, x_5 \in R$ and $|x_1 - x_2| = 2$, $|x_2 - x_3| = 4$, $|x_3 - x_4| = 3$, $|x_4 - x_5| = 5$.

Then the sun of all distinct possible values of $|x_5 - x_1|$ is

40. The number of integers which does NOT satisfy

$\log_{|2x|}(|x+2| + |x-2|) = 1$ is

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) 3

41. The real solutions of the equation where $|x|^2 - 3|x| + 2 = 0$ where $x_1 < x_2 < x_3 < x_4$ then

(*a*) $|x_1| = |x_3|$ (*b*) $|x_2| = |x_3|$

(*c*) $x_1 + x_4 = x_2 + x_3$ (*d*) $-x_2 + x_4 = x_1 - x_3$

42. Solve for *x*:

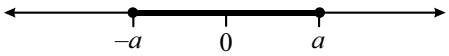
(*i*) $||x-2|-1| = 2$ (*ii*) $||x-3|-5| = 1$

(*iii*) $|||x-5|-4|-3| = 2$

INEQUALITIES INVOLVING ABSOLUTE VALUE

(*i*) $|x| \le a$ (where $a > 0$)

It implies those values of *x* on real number line which are at distance *a* or less than *a* from zero.



$\Rightarrow -a \le x \le a$

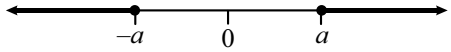
e.g. $|x| \le 2 \Rightarrow -2 \le x \le 2$

$|x| < 3 \Rightarrow -3 < x < 3$

In general, $|f(x)| \le a$ (where $a > 0$) $\Rightarrow -a \le f(x) \le a$.

(*ii*) $|x| \ge a$ (where $a > 0$)

It implies those values of *x* on real number line which are at distance *a* or more than *a* from zero



$\Rightarrow x \le -a$ or $x \ge a$

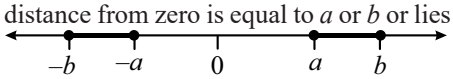
e.g. $|x| \ge 3 \Rightarrow x \le -3$ or $x \ge 3$

$|x| > 2 \Rightarrow x < -2$ or $x > 2$

In general, $|f(x)| \ge a \Rightarrow f(x) \le -a$ or $f(x) \ge a$

(*iii*) $a \le |x| \le b$ (where $a, b > 0$)

It implies those value of *x* on real number line whose distance from zero is equal to *a* or *b* or lies between *a* and *b*



$\Rightarrow [-b, -a] \cup [a, b]$

e.g. $2 \le |x| \le 4 \Rightarrow x \in [-4, -2] \cup [2, 4]$

(*iv*) If $|x+y| = |x| + |y|$, $xy \ge 0$

If $|x-y| = |x| + |y|$, $xy \le 0$

If $|x+y| = ||x| - |y||$, $xy \le 0$

If $|x-y| = ||x| - |y||$, $xy \ge 0$

Train Your Brain

Example 22: Solve $x^2 - 4|x| + 3 < 0$.

Sol. $x^2 - 4|x| + 3 < 0$

$\Rightarrow (|x| - 1)(|x| - 3) < 0$

$\Rightarrow 1 < |x| < 3$

$\Rightarrow -3 < x < -1$ or $1 < x < 3$

$\Rightarrow x \in (-3, -1) \cup (1, 3)$

Example 23: Solve $1 \le |x-2| \le 3$

Sol. $1 \le |x-2| \le 3$

$\Rightarrow -3 \le x-2 \le -1$ or $1 \le x-2 \le 3$

$\Rightarrow -1 \le x \le 1$ or $3 \le x \le 5$

$\Rightarrow x \in [-1, 1] \cup [3, 5]$

Example 24: Solve $|x-1| + |x-2| + |x-3| \ge 6$,

Sol. For $x \le 1$, the given inequation becomes

$1 - x + 2 - x + 3 - x \ge 6 \Rightarrow -3x \ge 0$

$\Rightarrow x \le 0$ and for $x \ge 3$, the given equation becomes

$x - 1 + x - 2 + x - 3 \ge 6 \Rightarrow 3x \ge 12 \Rightarrow x \ge 4$

For $1 < x \le 2$

we get $x - 1 + 2 - x + 3 - x \ge 6$

$\Rightarrow -x + 4 \ge 6$

i.e. $-x \ge 2 \Rightarrow x \le -2$ Not possible

For $2 < x < 3$,

We get $x - 1 + x - 2 + 3 - x \ge 6$

$\Rightarrow x \ge 6$ not possible

Hence solution set is $(-\infty, 0] \cup [4, \infty)$

i.e. $x \le 0$ or $x \ge 4$

Concept Application

43. Solve $||x-1|-2| < 5$

44. Number of non-positive integral values of '*x*' satisfying the given inequality, $|x^2 - 1| \le |2x - 1|$ is

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) 3

45. Solve $|x^2 - 2x| + |x - 4| > |x^2 - 3x + 4|$.

46. Solve $\left|\frac{x-3}{x+1}\right| \le 1$.

47. Solve for *x*:

(*i*) $|x| > 1$ (*ii*) $|x| \ge 5$ (*iii*) $|x| < 7$

(*iv*) $|x| \le 10$ (*v*) $|x| \ge 0$ (*vi*) $|x| < -8$

(*vii*) $|x| > -4$ (*viii*) $|x| \ge -5$ (*ix*) $|x| \le -10$

48. Solve for *x*:

(*i*) $|x-1| \ge 1$ (*ii*) $|x-2| < 1$

(*iii*) $1 < |2x+1| < 3$ (*iv*) $1 \le |1-2x| \le 3$

(*v*) $-1 \le |3x-1| \le 5$ (*vi*) $-6 \le |1-3x| \le -1$

49. Solve for *x*:

(*i*) $||x-2|-1| \le 2$ (*ii*) $||x-3|-5| \ge 1$

(*iii*) $|||x-5|-4|-3| \le 2$

Short Notes

Some Important Identities

1. $(a+b)^2 = a^2+2ab+b^2 = (a-b)^2+4ab$
2. $(a-b)^2 = a^2-2ab+b^2 = (a+b)^2-4ab$
3. $a^2-b^2 = (a+b)(a-b)$
4. $(a+b)^3 = a^3+b^3+3ab(a+b)$
5. $(a-b)^3 = a^3-b^3-3ab(a-b)$
6. $a^3+b^3 = (a+b)^3-3ab(a+b) = (a+b)(a^2+b^2-ab)$
7. $a^3-b^3 = (a-b)^3+3ab(a-b) = (a-b)(a^2+b^2+ab)$
8. $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca = a^2+b^2+c^2+2abc\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$
9. $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$
10. $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
    $= \frac{1}{2}(a+b+c)[(a-b)^2+(b-c)^2+(c-a)^2]$
    If $a+b+c=0$, then $a^3+b^3+c^3 = 3abc$
11. $a^4-b^4 = (a+b)(a-b)(a^2+b^2)$
12. $a^4+a^2+1 = (a^2+1)^2-a^2 = (1+a+a^2)(1-a+a^2)$

Laws of Indices

1. $a^m \times a^n = a^{m+n}$
2. $a^m \div a^n = a^{m-n}$
3. $(a^m)^n = (a^n)^m = a^{mn}$
4. $\left(\frac{a}{b}\right)^{\frac{m}{n}} = \left(\frac{b}{a}\right)^{-\frac{m}{n}}$
5. $a^m \div b^{-n} = a^m \times b^n$
6. $(\sqrt[n]{a})^n = a$, where, $n \in N$, $n \geq 2$ and a is positive rational number
7. $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$, where $n \in N$, $n \geq 2$ and a, b are rational number
8. $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$, $a, b \in R$ and atleast one of a or b should be positive.

Ratio and Proportion

a, b, c, d are in proportion. Then,

(*i*) $\frac{a}{b} = \frac{c}{d}$

(*ii*) $\frac{a}{c} = \frac{b}{d}$ .....(alternendo)

(*iii*) $\frac{a+b}{b} = \frac{c+d}{d}$ ....(componendo)

(*iv*) $\frac{a-b}{b} = \frac{c-d}{d}$ .....(dividendo)

(*v*) $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ .....(componendo and dividendo)

(*vi*) $\frac{b}{a} = \frac{d}{c}$ .....(invertendo)

(*vii*) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} =$ ....., then each

$$\frac{a+c+e+.....}{b+d+f+.....} = \frac{\text{Sum of the numerators}}{\text{Sum of the denominators}}$$

(*viii*) If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} =$....., then each $= \frac{xa+yc+ze+...}{xb+yd+zf+...}$

Solving of Inequalities by Wavy Curve Method

Step 1: Obtain critical points by equating all factors to zero.

Step 2: Plot the critical points on the number line in the increasing order.

Step 3: Put plus sign in the right most interval.

Step 4: Now, if a root is repeated even times, the sign of the function will remain the same in the two adjacent sub-intervals of the root. (when we are moving from right to left)

Step 5: If a root is repeated for odd times the sign of the function will be different in the two adjacent sub intervals of the root. (when we are moving from right to left)

Properties of Logarithm

1. $\log_e(ab) = \log_e a + \log_e b$; $(a, b > 0)$
2. $\log_e\left(\frac{a}{b}\right) = \log_e a - \log_e b$; $(a, b > 0)$
3. $\log_e a^m = m\log_e a$; $(a > 0, m \in R)$
4. $\log_a a = 1$; $(a > 0, a \neq 1)$
5. $\log_{b^m} a = \frac{1}{m}\log_b a$; $(a, b > 0, b \neq 1$ and $m \in R - \{0\})$
6. $\log_b a = \frac{1}{\log_a b}$; $(a, b, > 0$ and $a, b \neq 1)$
7. $\log_b a = \frac{\log_m a}{\log_m b}$; $(a, b, m > 0$ and $m, b \neq 1)$
8. $a^{\log_a m} = m$; $(a > 0, a \neq 1\ m > 0)$
9. $a^{\log_c b} = b^{\log_c a}$; $(a, b, c > 0$ and $c \neq 1)$
10. If $\log_m x > \log_m y \Rightarrow \begin{cases} x > y, if\ m > 1 \\ x < y, if\ 0 < m < 1 \end{cases}$
    $(m, x, y > 0, m \neq 1)$
11. $\log_m a = b \Rightarrow a = m^b$; $(m, a > 0, m \neq 1 \in$ real number$)$
12. $\log_m a > b \Rightarrow \begin{cases} a > m^b ; if\ m > 1 \\ a < m^b ; if\ 0 < m < 1 \end{cases}$

13. $\log_m a < b \Rightarrow \begin{cases} a < m^b ; if\ \ m > 1 \\ a > m^b ; if\ \ 0 < m < 1 \end{cases}$

Logarithmic Equations

(*i*) An equation of the form $\log_a f(x) = b, (a > 0), a \neq 1$ is equivalent to the equation. $f(x) = a^b$, $(f(x) > 0)$

(*ii*) If $\log_a f(x) > \log_a g(x)$ and $a > 1$, then $\Rightarrow \begin{cases} g(x) > 0 \\ f(x) > 0 \\ f(x) > g(x) \end{cases}$

(*iii*) If $\log_a f(x) > \log_a g(x)$ and $a < 1$, then $\Rightarrow \begin{cases} f(x) > 0 \\ g(x) > 0 \\ f(x) < g(x) \end{cases}$

Definition of Modulus

$$|x| = \begin{cases} x, \ if\ \ x \geq 0 \\ -x, \ \ if\ x < 0 \end{cases}$$

Properties of Modulus

Let '*a*', '*b*' are positive real number then,

(*i*) $|x| = a \Rightarrow x = \pm a$

(*ii*) $|x| \leq a \Rightarrow -a \leq x \leq a \Rightarrow x \in [-a, a]$

(*iii*) $|x| < a \Rightarrow -a < x < a \Rightarrow x \in (-a, a)$

(*iv*) $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a \Rightarrow x \in (-\infty, -a] \cup [a, \infty)$

(*v*) $|x| > a \Rightarrow x < -a$ or $x > a \Rightarrow x \in (-\infty, -a) \cup (a, \infty)$

(*vi*) $a \leq |x| \leq b \Rightarrow x \in [-b, -a] \cup [a, b]$

(*vii*) $a < |x| < b \Rightarrow x \in (-b, -a) \cup (a, b)$

(*viii*) $|x + y| \leq |x| + |y|$; equality holds when $xy \geq 0$

(*ix*) $|x - y| \geq ||x| - |y||$; equality holds when $xy \geq 0$

(*x*) $|x + y| \geq ||x| - |y||$; equality holds when $xy \leq 0$

(*xi*) $|x - y| \leq |x| + |y|$; equality holds when $xy \geq 0$

Solved Examples

1. The value of $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$ is equal to

(*a*) 49 (*b*) 625

(*c*) 216 (*d*) 890

Sol. (*d*) $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$

$= 3^{4\log_3 5} + 3^{3.\frac{1}{2}\log_3 36} + 3^{4\log_9 7}$

$= 3^{\log_3 5^4} + 3^{\log_3 36^{3/2}} + 3^{\log_3 7^{4/2}}$

$= 5^4 + 36^{3/2} + 7^2 = 890$

2. Values of x satisfying the equation

$\log_5^2 x + \log_{5x}\left(\frac{5}{x}\right) = 1$ are

(*a*) 1 (*b*) 5

(*c*) $\frac{1}{25}$ (*d*) 3

Sol. (*a*,*b*,*c*)

$(\log_5 x)^2 + \log_{5x} \frac{5}{x} = 1$

$\Rightarrow (\log_5 x)^2 + \log_{5x} 5 - \log_{5x} x = 1$

$\Rightarrow (\log_5 x)^2 + \frac{\log_5 5}{\log_5 5 + \log_5 x} - \frac{\log_5 x}{\log_5 5 + \log_5 x} = 1$

$\Rightarrow (\log_5 x)^2 + \frac{1}{1 + \log_5 x} - \frac{\log_5 x}{1 + \log_5 x} = 1$

Let $\log_5 x = t$

$\therefore\ t^2 + \frac{1}{1+t} - \frac{t}{1+t} = 1$

$\Rightarrow \frac{t^2(1+t) + 1 - t}{1+t} = 1$

$\Rightarrow t^3 + t^2 + 1 - t = 1 + t$

$t^3 + t^2 - 2t = 0$

$t(t^2 + t - 2) = 0$

$t(t - 1)(t + 2) = 0$

$t = 0, 1, -2$

$\therefore \log_5 x = 0, 1, -2$

$\therefore x = 1, 5, \frac{1}{25}$

3. The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has

(*a*) One irrational solution

(*b*) No prime solution

(*c*) Two real solutions

(*d*) One integral solution

Sol. (*a*,*b*,*c*,*d*)

$\frac{4}{2}\log_x 2 + \frac{\log_x 64}{\log_x 2x}$

$\Rightarrow \frac{2x\log_x 2}{x} + \frac{6\log_x 2}{1 + \log_x 2} = 3$

Let $\alpha = \log_x 2$

$2\alpha + \frac{6\alpha}{1+\alpha} = 3$

$2\alpha + 2\alpha^2 + 6\alpha - 3 - 3\alpha = 0$

$\Rightarrow 2\alpha^2 + 5\alpha - 3 = 0$

$\Rightarrow (\alpha + 3)(2\alpha - 1) = 0 \Rightarrow a = -3, 1/2$

$\therefore \log_x 2 = -3 \Rightarrow x = 2^{-1/3}$ (Irrational)

or $\log_x 2 = \frac{1}{2} \Rightarrow x = 4$ (Integer)

4. If $\frac{1}{2} \le \log_{0.1} x \le 2$, then

(*a*) Maximum value of x is $\frac{1}{\sqrt{10}}$

(*b*) x lies between $\frac{1}{100}$ and $\frac{1}{\sqrt{10}}$

(*c*) Minimum value of x is $\frac{1}{10}$

(*d*) Minimum value of x is $\frac{1}{100}$

Sol. (*a*,*b*,*d*)

$$\frac{1}{2} \le \log_{0.1} x \le 2 \Rightarrow \left(\frac{1}{10}\right)^{1/2} \ge x \ge \left(\frac{1}{10}\right)^2$$

5. Let $N = \frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$. Then N is

(*a*) a natural number (*b*) a prime number

(*c*) a rational number (*d*) an integer

Sol. (*a*,*b*,*c*,*d*)

$= \log_3 135 \log_3 15 - \log_3 5 \log_3 405$

$= \log_3(5 \times 3^3) \cdot \log(5 \times 3) - \log_3 5. \log_3 (5 \times 3^4)$

$= (\log_3 5 + \log_3 3^3)(\log_3 5 + \log_3 3) - \log_3 5 (\log_3 5 + \log_3 3^4)$

$= (x + 3)(x + 1) - x(x + 4)$

{Let $\log_3 5 = x$}

$= x^2 + 4x + 3 - x^2 - 4x = 3$

which is Prime, rational Integer and natural number

6. If $|x - 5| + |x + 5| = 10$, then

(*a*) The number of integral solutions is 10

(*b*) The number of integral solutions is 11

(*c*) The sum of all the integral solutions is 0

(*d*) All the solutions of the equation are rational numbers

Sol. (*b*,*c*)

$|x - 5| + |x + 5| = 10$

Case-I: $x \ge 5$, the equation becomes

$(x - 5) + (x + 5) = 10$

$\Rightarrow 2x = 10$

$\Rightarrow x = 5$ which satisfies the case, therefore accepted.

Case-II: $-5 < x < 5$ The above equation becomes

$-(x - 5) + (x + 5) = 10$

$\Rightarrow -x + 5 + x + 5 = 10$

$\Rightarrow 10 = 10$ which is true.

So, the solution is $x \in (-5, 5)$

Case-III: $x \le -5$, The above equation becomes

$-(x - 5) - (x + 5) = 10$

$\Rightarrow -x + 5 - x - 5 = 10$

$\Rightarrow -2x = 10$

$\Rightarrow x = -5$ which satisfies the above case so, accepted.

$\therefore$ final answer is $x \in [-5, 5]$

7. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10.$

Find $y \in N$.

Sol. [0006]

$$y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$$

$$= \sqrt{\log_2 3 \cdot (2 + \log_2 3)(4 + \log_2 3)(6 + \log_2 3) + 16} - (2 + \log_2 3)(4 + \log_2 3) + 10$$

Let us put $\log_2 3 = x$

$$= \sqrt{x(2 + x)(4 + x)(6 + x) + 16} - (2 + x)(4 + x) + 10$$

$$= \sqrt{(x^2 + 6x)(x^2 + 6x + 8) + 16} - (x^2 + 6x + 8) + 10$$

Put again $x^2 + 6x = \alpha$

$$= \sqrt{\alpha(\alpha + 8) + 16} - (\alpha + 8) + 10$$

$$= \sqrt{\alpha^2 + 8\alpha + 16} - (\alpha + 8) + 10$$

$$= \sqrt{(\alpha + 4)^2} - (\alpha + 8) + 10$$

$$= (\alpha + 4) - (\alpha + 8) + 10 = y = 6.$$

8. If 'x' and 'y' are real numbers such that,

$2 \log(2y - 3x) = \log x + \log y$, find $\frac{y}{x}$.

Sol. [2.25]

$\log(2y - 3x)^2 = \log xy$

$(2y - 3x)^2 = xy$

$4y^2 - 12xy + 9x^2 = xy$

Dividing the equation by y^2

$$9\left(\frac{x}{y}\right)^2 - 13\frac{x}{y} + 4 = 0$$

$$\left(\frac{x}{y} - 1\right)\left(\frac{9x}{y} - 4\right) = 0$$

$$\frac{x}{y} = 1, \frac{x}{y} = \frac{4}{9}.$$

$x = y$ disregarded as for $x = y$, $2y - 3x$ is negative.

Hence $\frac{y}{x} = \frac{9}{4}$.

9. Sum of all the solutions of the equation

$\log_6(x^2 - 1) - \log_6 \sqrt{(x - 6)^2} = \log_6 (x + 1)^2$ is $a + \sqrt{b}$, $(a, b \in N)$. Then $a + b$ is equal to

Sol. $\log_6(x^2 - 1) - \log_6 \sqrt{(x - 6)^2} = \log_6 (x + 1)^2$

$$\Rightarrow \log_6 \frac{(x - 1)(x + 1)}{(x + 1)^2} = \log_6 |x - 6|$$

$\Rightarrow \log_6\left[\frac{(x-1)}{(x+1)}\right] = \log_6 |x-6|$

$\Rightarrow \frac{x-1}{x+1} = |x-6|$

Case-I: $x \geq 6$

$\Rightarrow x - 1 = x^2 - 5x - 6$

$\Rightarrow x^2 - 6x - 5 = 0$

$\Rightarrow (x-3)^2 = 14$

$\Rightarrow x = 3 \pm \sqrt{14}$

$x = 3 - \sqrt{14} < 1$ rejected

$x = 3 + \sqrt{14}$ accepted

Case-II: $x < 6$

$x - 1 = -(x^2 - 5x - 6)$

$\Rightarrow x^2 - 4x - 7 = 0$

$(x-2)^2 = 11$

$x = 2 \pm \sqrt{11}$

$x = 2 + \sqrt{11}$ (accepted)

$x = 2 - \sqrt{11}$ (accepted)

Sum of roots = $7 + \sqrt{14}$

$\Rightarrow a = 7, b = 14$

$a + b = 21$

10. Match the column

Column-I		Column-II	
A.	The roots of $\log_2(x+e) = \log_2 x + \log_2 e$ is a	p.	Positive Number
B.	The solution of $\log_{1/5}(2x^2 + 5x + 1) < 0$ contains	q.	Rational Number
C.	$\log_{\sin\frac{\pi}{6}} \pi$ is	r.	Irrational Number
D.	$\log_{10}5.\log_{10}20 + \log_{10}^2 2$ simplifies to	s.	Negative Number

Sol. A→(p, r) ; B→(p, q, r, s) ; C→(r, s) ; D→(p, q)

(*A*) $x + e = xe$

$x(e-1) = e$

$x = \frac{e}{e-1}$

(*B*) $2x^2 + 5x + 1 > 1$ and $2x^2 + 5x + 1 > 0$

$\Rightarrow 2x^2 + 5x + 1 > 1$

$\Rightarrow (x)(2x + 5) > 0$

$\Rightarrow x \in \left(-\infty, \frac{-5}{2}\right) \cup (0, \infty)$

(*D*) $(1 - \log_{10} 2)(1 + \log_{10} 2) + \log_{10}^2 2$

$\Rightarrow 1 - \log_{10}^2 2 + \log_{10}^2 2 = 1$

11. The largest integral value of x satisfying

$\sqrt{18^x - 5} \leq \sqrt{2(18^x + 12)} - \sqrt{18^x + 5}$ is

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) no integral value of x possible

Sol. (*d*) Let $18^x = p$

$\sqrt{p-5} + \sqrt{p+5} \leq \sqrt{2(p+12)}$

$\Rightarrow p - 5 + p + 5 + 2\sqrt{p^2 - 25} \leq 2p + 24$

$\Rightarrow \sqrt{p^2 - 25} \leq 12 \Rightarrow p^2 \leq 169 \Rightarrow p \leq 13$

Also $p \geq 5$

Thus $5 \leq p \leq 13$ $\log_{18}5 \leq x \leq \log_{18}13$

12. If $\log_a x = b$ for permissible values of a and x then identify the statements(*s*) which can be correct?

(*a*) If a and b are two irrational numbers then x can be rational.

(*b*) If a rational and b irrational then x can be rational.

(*c*) If a irrational and b rational then x can be rational.

(*d*) If a rational and b rational then x can be rational.

Sol. (*a*,*b*,*c*,*d*)

(*a*) $a = (\sqrt{2})^{\sqrt{2}}$ is irrational

$b = \sqrt{2}$ is also irrational

but $a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = 2$ which is rational $\Rightarrow$ (*a*) is correct.

(*b*) $a = 2 \in Q$; $b = \log_2 3 \notin Q$

$a^b = 2^{\log_2 3} = 3 \in Q \Rightarrow$ (*b*) is correct

13. Solve if $|x-5| + |x+4| = 9$

Sol. Given equation is of form $|a| + |b| = |a - b|$

It is true for $ab \leq 0$

$(x-5)(x+4) \leq 0$

So $x \in [-4, 5]$

14. Solve $\frac{(e - \sin x)(x-2)}{(x+4)} \geq 0$.

Sol. Zeroes $x = 2$, Pole $x \neq -4$

$e - \sin x > 0$ always positive

$\frac{(e - \sin x)(x-2)}{(x+4)} \geq 0$

Final solution $x \in (-\infty, -4) \cup [2, \infty)$

Exercise-1 (Topicwise)

BASIC CONCEPTS AND NUMBER SYSTEM

1. Let $x \in Q$, $y \in Q^c$, Which of the following statement is always WRONG?

(*a*) $xy \in Q^c$

(*b*) $y/x \in Q$, whenever defined

(*c*) $\sqrt{2}\,x + y \in Q$

(*d*) $x/y \in Q^c$, whenever defined

2. If x and y are two rational numbers such that $(x+y)+(x-2y)\sqrt{2} = 2x - y + (x - y - 1)\sqrt{6}$, then:

(*a*) $x = 1, y = 1$

(*b*) $x = 2, y = 1$

(*c*) $x = 5, y = 1$

(*d*) x and y can take infinitely many values

3. Which of the following statement is incorrect :

(*a*) rational number + rational number = rational number

(*b*) irrational number + rational number = irrational number

(*c*) integer + rational number = rational number

(*d*) irrational number + irrational number = Irrational number

4. The number of real roots of the equation $(x-1)^2 + (x-2)^2 + (x-3)^2 = 0$ is :

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) 3

5. If $x - a$ is a factor of $x^3 - a^2x + x + 2$, then 'a' is equal to

(*a*) 0 (*b*) 2

(*c*) –2 (*d*) 1

6. Every irrational number can be expressed on the number line. This statement is

(*a*) Always true

(*b*) Never true

(*c*) True subject to some condition

(*d*) None of these

7. The multiplication of a rational number 'x' and an irrational number 'y' is

(*a*) Always rational

(*b*) Rational except when $y = \pi$

(*c*) Always irrational

(*d*) Irrational except when $x = 0$

8. If x, y are integral solutions of $2x^2 - 3xy - 2y^2 = 7$, then value of $|x + y|$ is

(*a*) 2 (*b*) 4

(*c*) 6 (*d*) 2 or 4 or 6

9. If a, b, c are real, then $a(a-b) + b(b-c) + c(c-a) = 0$, only if

(*a*) $a + b + c = 0$

(*b*) $a = b = c$

(*c*) $a = b$ or $b = c$ or $c = a$

(*d*) $a - b - c = 0$

10. If $2x^3 - 5x^2 + x + 2 = (x-2)(ax^2 - bx - 1)$, then a & b are respectively

(*a*) 2, 1 (*b*) 2, – 1

(*c*) 1, 2 (*d*) –1, 1/2

11. The value of $[e] - [-\pi]$ is, where [.] denotes greatest integer function.

(*a*) 5 (*b*) 6

(*c*) 7 (*d*) 8

12. If $L = \frac{1}{\sqrt{7}-\sqrt{8}} + \frac{1}{\sqrt{7}-\sqrt{6}} + \frac{1}{3-\sqrt{8}} + \frac{1}{\sqrt{5}+2} + \frac{1}{\sqrt{5}-\sqrt{6}}$ $= 1 + 2\sqrt{a} + 2\sqrt{b}$, then $a.\ b$ is equal to

(*a*) 30 (*b*) 45

(*c*) 8 (*d*) 16

13. If a, b, c are real and distinct numbers, then the value of $\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$ is

(*a*) 1 (*b*) $a\,b\,c$

(*c*) 2 (*d*) 3

14. The remainder obtained when the polynomial $1 + x + x^3 + x^9 + x^{27} + x^{81} + x^{243}$ is divided by $x - 1$ is

(*a*) 3 (*b*) 5

(*c*) 7 (*d*) 11

15. $\frac{1}{1+\log_b a + \log_b c} + \frac{1}{1+\log_c a + \log_c b} + \frac{1}{1+\log_a b + \log_a c}$ has the value equal to

(*a*) abc (*b*) $\frac{1}{abc}$

(*c*) 0 (*d*) 1

16. $\log_7 \log_7 \sqrt{7(\sqrt{7\sqrt{7}})} =$

(*a*) $3 \log_2 7$ (*b*) $1 - 3 \log_3 7$

(*c*) $1 - 3\log_7 2$ (*d*) $1 - 10 \log_2 7$

LOGARITHM AND ITS PRINCIPLE PROPERTIES

17. $\frac{1}{\log_{\sqrt{bc}} abc}+\frac{1}{\log_{\sqrt{ca}} abc}+\frac{1}{\log_{\sqrt{ab}} abc}$ has the value equal to

(*a*) 1/2 (*b*) 1

(*c*) 2 (*d*) 4

18. If $\log_x \log_{18}(\sqrt{2}+\sqrt{8})=\frac{1}{3}$. Then the value of 1000 x is equal to

(*a*) 8 (*b*) 1/8

(*c*) 1/125 (*d*) 125

19. Number of real solutions of the equation $\sqrt{\log_{10}(-x)}=\log_{10}\sqrt{x^2}$ is :

(*a*) none (*b*) exactly 1

(*c*) exactly 2 (*d*) 4

20. Greatest integer less than or equal to the number $\log_2 15.\log_{1/6} 2.\log_3 1/6$ is

(*a*) 4 (*b*) 3

(*c*) 2 (*d*) 1

21. The ratio $\frac{2^{\log_{2^{1/4}} a}-3^{\log_{27}(a^2+1)^3}-2a}{7^{4\log_{49} a}-a-1}$ simplifies to

(*a*) a^2-a-1 (*b*) a^2+a-1

(*c*) a^2-a+1 (*d*) a^2+a+1

22. If $3^{2\log_3 x}-2x-3=0$, then the number of values of 'x' satisfying the equation is

(*a*) zero (*b*) 1

(*c*) 2 (*d*) More than 2

23. The number $\log_2 7$ is

(*a*) an integer (*b*) a rational number

(*c*) an irrational number (*d*) a prime number

24. Anti logarithm of 0.75 to the base 16 has the value equal to

(*a*) 4 (*b*) 6

(*c*) 8 (*d*) 12

25. The sum of all the solutions to the equation $2\log x - \log(2x-75)=2$:

(*a*) 30 (*b*) 350

(*c*) 75 (*d*) 200

INEQUALITIES

26. If the solution set of the inequality $\log_{\sqrt{0.9}} \log_5(\sqrt{x^2+5+x})>0$ contains 'n' integral values, then n equals to

(*a*) 7 (*b*) 8 (*c*) 6 (*d*) 10

27. If $\log_{0.5}\log_5(x^2-4)>\log_{0.5}1$, then '$x$' lies in the interval

(*a*) $(-3,-\sqrt{5})\cup(\sqrt{5},3)$ (*b*) $(-3,-\sqrt{5})\cup(\sqrt{5},3\sqrt{5})$

(*c*) $(\sqrt{5},3\sqrt{5})$ (*d*) ϕ

28. Solution set of the inequality $2-\log_2(x^2+3x)\geq 0$ is:

(*a*) $[-4,1]$ (*b*) $[-4,-3)\cup(0,1]$

(*c*) $(-\infty,-3)\cup(1,\infty)$ (*d*) $(-\infty,-4)\cup[1,\infty)$

MODULUS FUNCTION

29. Solutions of $|4x+3|+|3x-4|=12$ are

(*a*) $x=-\frac{7}{3},\frac{3}{7}$ (*b*) $x=-\frac{5}{2},\frac{2}{5}$

(*c*) $x=-\frac{11}{7},\frac{13}{7}$ (*d*) $x=-\frac{3}{7},\frac{7}{5}$

30. If $|x^2-2x-8|+|x^2+x-2|=3|x+2|$, then the set of all real values of x is

(*a*) $[1,4]\cup\{-2\}$ (*b*) $[1,4]$

(*c*) $[-2,1]\cup[4,\infty)$ (*d*) $(-\infty,-2]\cup[1,4]$

31. The complete set of real 'x' satisfying $||x-1|-1|\leq 1$ is:

(*a*) $[0,2]$ (*b*) $[-1,3]$

(*c*) $[-1,1]$ (*d*) $[1,3]$

32. The number of real roots of the equation $|x|^2-3|x|+2=0$ is

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

33. Number of real solution (x) of the equation $|x-3|^{3x^2-10x+3}=1$ is

(*a*) exactly four (*b*) exactly three

(*c*) exactly two (*d*) exactly one

34. Simplify: $7^{\log_3 5}+3^{\log_5 7}-5^{\log_3 7}-7^{\log_5 3}$

(*a*) 0 (*b*) 1 (*c*) 3 (*d*) 4

35. The expression $x^2-y^2-z^2+2yz+x+y-z$ has a factor

(*a*) $x-y+z+1$ (*b*) $-x+y+z$

(*c*) $x+y-z+1$ (*d*) $x-y-z+1$

36. Solve the equation $\frac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3}=\frac{5x^4+2x^2-7x+3}{5x^4-2x^2+7x-3}$

(*a*) $x=5,2$ (*b*) $x=4,1$

(*c*) $x=3,8$ (*d*) $x=1,5$

37. If x, y, z are positive real number and a, b, c are rational numbers, then the value of $\frac{1}{1+x^{b-a}+x^{c-a}}+\frac{1}{1+x^{a-b}+x^{c-b}}+\frac{1}{1+x^{b-c}+x^{a-c}}$ is

(*a*) –1 (*b*) 1 (*c*) 0 (*d*) 2

38. If $a^x=\sqrt{b}, b^y=\sqrt[3]{c}$ and $c^z=\sqrt{a}$ then the value of xyz is

(*a*) $\frac{1}{2}$ (*b*) $\frac{1}{3}$ (*c*) $\frac{1}{6}$ (*d*) $\frac{1}{12}$

39. If $\frac{\log a}{b-c}=\frac{\log b}{c-a}=\frac{\log c}{a-b}$, then $a^a \cdot b^b \cdot c^c =$

(*a*) 3 (*b*) 1 (*c*) 4 (*d*) 2

40. The number of prime numbers satisfying the inequality $\frac{x^2-1}{2x+5}<3$ is

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

Exercise-2 (Learning Plus)

1. If A & B are two rational numbers and AB, A + B and A – B are rational numbers, then A/B is
(*a*) Always rational (*b*) Never rational
(*c*) Rational when B ≠ 0 (*d*) Rational when A ≠ 0

2. If $x^{x\sqrt[3]{x}} = \left(x.\sqrt[3]{x}\right)^x$, then x =
(*a*) 1 (*b*) –1
(*c*) 0 (*d*) 2

3. The equation $4^{\left(x^2+2\right)} - 9.2^{\left(x^2+2\right)} + 8 = 0$ has the solution
(*a*) $x = \pm 1$ (*b*) $x = 10$
(*c*) $x = \pm\sqrt{2}$ (*d*) $x = \sqrt{3}$

4. Logarithm of $32\sqrt[5]{4}$ to the base $2\sqrt{2}$ is
(*a*) 3.6 (*b*) 5
(*c*) 5.6 (*d*) 10

5. If $x = \log_a(bc)$, $y = \log_b(ca)$, $z = \log_c(ab)$, then which of the following is equal to 1
(*a*) $x + y + z$
(*b*) $(1 + x)^{-1} + (1 + y)^{-1} + (1 + z)^{-1}$
(*c*) xyz
(*d*) $x + y - z$

6. The solution of the equation $\log_7 \log_5 \left(\sqrt{x^2+5}+x\right) = 0$.
(*a*) $x = 2$ (*b*) $x = 3$
(*c*) $x = 4$ (*d*) $x = -2$

7. The value of $(0.05)^{\log_{\sqrt{20}}(0.1+0.01+0.001+.....)}$ is
(*a*) 81 (*b*) $\frac{1}{81}$
(*c*) 20 (*d*) $\frac{1}{20}$

8. The value of $\log_2 .\log_3 ....\log_{100} 100^{99^{98^{\cdot^{\cdot^{2^1}}}}}$ is
(*a*) 0 (*b*) 1
(*c*) 2 (*d*) 100!

9. The number of solution of $\log_2(x + 5) = 6 - x$ is
(*a*) 2 (*b*) 0
(*c*) 3 (*d*) 1

10. If $\log_{10}2 = 0.30103$, $\log_{10}3 = 0.47712$, the number of digits in 312×28 is
(*a*) 7 (*b*) 8
(*c*) 9 (*d*) 10

11. Exhaustive set of values of x satisfying $\log_{|x|}(x2 + x + 1) \geq 0$ is
(*a*) (–1, 0) (*b*) $(-\infty, -1) \cup (1, \infty)$
(*c*) $(-\infty, \infty) - \{-1, 0, 1\}$ (*d*) $(-\infty, -1) \cup (-1, 0) \cup (1, \infty)$

12. The set of real values of x satisfying $\log_{1/2}\left(x^2 - 6x + 12\right) \geq -2$ is
(*a*) $(-\infty, 2]$ (*b*) [2, 4]
(*c*) $[4, +\infty)$ (*d*) [3, 8]

13. If $\log_{0.04}(x - 1) \geq \log_{0.2}(x - 1)$ then x belongs to the interval
(*a*) (1, 2] (*b*) $(-\infty, 2]$
(*c*) [2, + ∞) (*d*) [2, 2]

14. If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval
(*a*) (2, ∞) (*b*) (– 2, –1)
(*c*) (1, 2) (*d*) (–2, 2)

15. The minimum value of $f(x) = |x - 1| + |x - 2| + |x - 3|$ is equal to
(*a*) 1 (*b*) 2
(*c*) 3 (*d*) 0

16. The set of real value(s) of p for which the equation $|2x + 3| + |2x - 3| = px + 6$ has more than two solutions is :
(*a*) [0, 4) (*b*) (– 4, 4)
(*c*) R – {4, – 4, 0} (*d*) {0}

17. Let $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$. The value of the product ($abcdef$) is :
(*a*) 1 (*b*) 2
(*c*) $\sqrt{6}$ (*d*) 3

18. There are two positive solutions to the equation $\log_{2x}2 + \log_4 2x = -\frac{3}{2}$. The product of these two solution is:
(*a*) $\frac{1}{32}$ (*b*) $\frac{1}{8}$ (*c*) $\frac{1}{2}$ (*d*) $\frac{1}{21}$

19. Number of real value of x satisfying the equation $\log_2(2x^2 + \sqrt{2}) = \frac{\sqrt{x^2+1}}{x^2+2}$ is :
(*a*) 0 (*b*) 1 (*c*) 2 (*d*) 3

20. Number of values of x satisfying the equation $x^4 = |x|^{\log_2(x^2+12)}$ is:
(*a*) 2 (*b*) 3
(*c*) 4 (*d*) 5

21. The number of zeros after decimal before the start of any significant digit in the number $N = (0.15)^{20}$ are :
(*a*) 15 (*b*) 16
(*c*) 17 (*d*) 18

22. If $n, k \in N$, then the smallest value of k such that $k + 2k + 3k \ldots 24k = n^3$ is

(*a*) 100 (*b*) 90

(*c*) 120 (*d*) 60

23. If $a, b, c \in R$ and $a, b, c \neq 0$ such that $\frac{a}{b}+\frac{b}{c}+\frac{c}{a}=6$ and $\frac{b}{a}+\frac{c}{b}+\frac{a}{c}=8$ then $\frac{a^3}{b^3}+\frac{b^3}{c^3}+\frac{c^3}{a^3}-3$ is equal to

(*a*) 81 (*b*) 48

(*c*) 72 (*d*) 84

24. If $3^x = 5^y$, then how many ordered pairs $(x, y), x, y \in R$ satisfy the given equation?

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) Infinite

25. The value of $x + y + z$ satisfying the system of equations

$\log_2 x + \log_4 y + \log_4 z = 2$ is

$\log_3 y + \log_9 z + \log_9 x = 2$

$\log_4 z + \log_{16} x + \log_{16} y = 2$

(*a*) $\frac{175}{12}$ (*b*) $\frac{349}{24}$

(*c*) $\frac{353}{24}$ (*d*) $\frac{112}{3}$

26. Which is the correct order for a given number α, $\alpha > 1$

(*a*) $\log_2 \alpha < \log_3 \alpha < \log_e \alpha > \log_{10} \alpha$

(*b*) $\log_{10} \alpha < \log_3 \alpha < \log_e \alpha > \log_2 \alpha$

(*c*) $\log_{10} \alpha < \log_e \alpha < \log_2 \alpha > \log_3 \alpha$

(*d*) $\log_3 \alpha < \log_e \alpha < \log_2 \alpha > \log_{10} \alpha$

27. The smallest integral value of x such that $\sqrt{x+2}-\sqrt{x-2}<\frac{1}{10}$ is

(*a*) 400 (*b*) 20

(*c*) 401 (*d*) 399

28. The ratio $\frac{2^{\log_{2^{1/4}} a}-3^{\log_{27}(a^2+1)^3}-2a}{7^{4\log_{49} a}-a-1}$ simplifies to

(*a*) $a^2 - a - 1$ (*b*) $a^2 + a - 1$

(*c*) $a^2 - a + 1$ (*d*) $a^2 + a + 1$

29. $10^{\log_p(\log_q(\log_r x))}=1$ and $\log_q(\log_r(\log_p x))=0$ then 'p' equals

(*a*) $r^{q/r}$ (*b*) rq

(*c*) 1 (*d*) $r^{r/q}$

30. Which one of the following is the smallest?

(*a*) $\log_{10}\pi$ (*b*) $\sqrt{\log_{10}\pi^2}$

(*c*) $\left(\frac{1}{\log_{10}\pi}\right)^3$ (*d*) $\left(\frac{1}{\log_{10}\sqrt{\pi}}\right)$

31. The set of all solutions of the inequality $(1/2)^{x^2-2x} < 1/4$ contains the set

(*a*) $(-\infty, 0)$ (*b*) $(-\infty, 1)$

(*c*) $(1, \infty)$ (*d*) $(3, \infty)$

32. The value of b satisfying thee quation, $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10$ is

(*a*) 5 (*b*) 7

(*c*) 9 (*d*) 10

33. The solution set of the system of equation $\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x + y) = \frac{2}{3}$ is:

(*a*) {6, 3} (*b*) {9, 6}

(*c*) {6, 12} (*d*) {12, 6}

34 Consider the quadratic equation, $(\log_{10}8)x^2 - (\log_{10}5)x = 2(\log_2 10)^{-1} - x$. Which of the following quantities are irrational.

(*a*) Sum of the roots (*b*) Product of the roots

(*c*) Sum of the coefficients (*d*) Discriminant

35. Which of the following statements are true?

(*a*) $\log_2 3 < \log_{12} 10$ (*b*) $\log_6 5 < \log_7 8$

(*c*) $\log_3 26 > \log_2 9$ (*d*) $\log_{16} 15 > \log_{10} 11 > \log_7 6$

36. If $x + y = a$ and $x^2 + y^2 = b$, then the value of $(x^3 + y^3)$ is

(*a*) ab (*b*) $a^2 + b$

(*c*) $a + b^2$ (*d*) $\frac{3ab - a^3}{2}$

37. If $2x^3 - 5x^2 + x + 2 = (x - 2)(ax^2 - bx - 1)$, then a & b are respectively

(*a*) 2, 1 (*b*) 2, –1 (*c*) 1, 2 (*d*) –1, 1/2

38. If $x + y + z = 0$, then a factor of the expression $(x + y)^3 + (y + z)^3 + (z + x)^3$ is

(*a*) $3(x + y)(y + z)(z + x)$ (*b*) $3xyz$

(*c*) $(x + y - z)$ (*d*) $(x - y + z)$

39. The number of real solution/s of the equation $9^{\log_3(\log_e x)} = \log_e x - (\log_e x)^2 + 1$ is:

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) 3

40. The set of all the solutions of the inequality $\log_{1-x}(x - 2) \geq -1$ is

(*a*) $(-\infty, 0)$ (*b*) $(2, \infty)$

(*c*) $(-\infty, 1)$ (*d*) ϕ

41. The complete solution of $\frac{x^2-1}{x+3}\geq 0 \;\&\; x^2-5x+2\leq 0$ is:

(*a*) $x \in \left[\frac{5-\sqrt{17}}{2}, \frac{5+\sqrt{17}}{2}\right]$ (*b*) $x \in \left[1, \frac{5-\sqrt{17}}{2}\right]$

(*c*) $x \in (-3, -1]$ (*d*) $x \in (-3, -1]\,[1, \infty)$

42. The number of the integral solutions of $x^2 + 9 < (x + 3)^2 < 8x + 25$ is:

(*a*) 1 (*b*) 3 (*c*) 4 (*d*) 5

43. Number of non-negative integral values of x satisfying the inequality $\frac{2}{x^2-x+1}-\frac{1}{x+1}-\frac{2x-1}{x^3+1}\geq 0$ is

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) 3

44. Match the values of x given in Column-II satisfying the exponential equation given in Column-I (Do not verify). Remember that for $a > 0$, the terms a^x is always greater than zero $\forall\ x \in R$.

Column-I		Column-II	
A.	$5^x - 24 = \frac{25}{5^x}$	p.	–3
B.	$(2^{x+1})(5^x) = 200$	q.	–2
C.	$4^{2/x} - 5(4^{1/x}) + 4 = 0$	r.	–1
D.	$2^{2x+1} - 33(2^{x-1}) + 4 = 0$	s.	0
E.	$\frac{2^{x-1}\cdot 4^{x+1}}{8^{x-1}} = 16$	t.	1
F.	$3^{2x+1} + 10\,(3^x) + 3 = 0$	u.	2
G.	$64(9^x) - 84(12^x) + 27(16^x) = 0$	v.	3
H.	$5^{2x} - 7^x - 5^{2x}(35) + 7^x(35) = 0$	w.	None

(*a*) $A \to u, B \to u, C \to t, D \to q, v, C \to p,q,r,s,t,u,v$ $F \to w, G \to t,u, H \to s$

(*b*) $A \to q, B \to t, C \to u, D \to v, C \to t,u,v\ F \to w,$ $G \to t, H \to q$

(*c*) $A \to p, B \to u, C \to s, D \to q, C \to p,q, F \to s, G \to u,$ $H \to q$

(*d*) $A \to q, B \to s, C \to r, D \to s, C \to q,r, F \to q, G \to u,$ $H \to q$

45. Match the columns:

Column-I		Column-II	
A.	$\log_{\sin 30^\circ}(\cos 60^\circ) + 1$	p.	3
B.	$\log_{4/3}(1.\overline{3}) + 3$	q.	5
C.	$\log_{2-\sqrt{3}}(2+\sqrt{3}) + 6$	r.	4
D.	$\log_{\tan 20^\circ} \tan 70^\circ + 4$	s.	2
E.	$\log_{\cot 40^\circ} \tan 50^\circ$	t.	0
F.	$\log_{0.125}(8) + 8$	u.	–1
G.	$\log_{1.5}(0.\overline{6}) + 9$	v.	8
H.	$\log_{2.25}(0.\overline{4})$	w.	7
I.	$\log_{10}(0.\overline{9})$	x.	1

(*a*) $A \to q, B \to p, C \to s, D \to v, E \to u, F \to u, G \to q,$ $H \to w, I \to x$

(*b*) $A \to s, B \to r, C \to q, D \to p, E \to t, F \to u, G \to v,$ $H \to w, I \to x$

(*c*) $A \to s, B \to v, C \to t, D \to p, E \to t, F \to u, G \to w,$ $H \to x, I \to w$

(*d*) $A \to q, B \to s, C \to r, D \to v, E \to u, F \to v, G \to v, H$ $\to w, I \to x$

46. Match the following columns.

Column-I		Column-II	
A.	If $a = 3\left(\sqrt{8+2\sqrt{7}} - \sqrt{8-2\sqrt{7}}\right)$, $b = \sqrt{(42)(30)+36}$ then the value of $\log_a b$ is equal to	p.	–1
B.	If $a = \sqrt{4+2\sqrt{3}} - \sqrt{4-2\sqrt{3}}$, $b = \sqrt{(42)(30)+36}$ then the value of $\log_a b$ is equal to	q.	1
C.	If $a = \sqrt{3+2\sqrt{2}}, b = \sqrt{3-2\sqrt{2}}$ then the value of $\log_a b$ is equal to	r.	2
D.	If $a = \sqrt{7+\sqrt{7^2-1}}, b = \sqrt{7-\sqrt{7^2-1}}$, then the value of $\log_a b$ is equal to	s.	$2 + 2\log_2 3$

(*a*) $A \to s, B \to p, C \to q, D \to p$

(*b*) $A \to r, B \to p, C \to r, D \to p$

(*c*) $A \to r, B \to s, C \to p, D \to p$

(*d*) $A \to p, B \to q, C \to p, D \to r$

47. Suppose $x, y, z > 0$ and different than one and $\ln x + \ln y + \ln z = 0$. Find the value of $K = x^{\frac{1}{\ln y}+\frac{1}{\ln z}} \cdot y^{\frac{1}{\ln z}+\frac{1}{\ln x}} \cdot z^{\frac{1}{\ln x}+\frac{1}{\ln y}}$.

48. If $\log_2(\log_3(\log_4(x))) = 0$ and $\log_3(\log_4(\log_2(y))) = 0$ and $\log_4(\log_2(\log_3(z))) = 0$ then find the sum of x, y and z is

49. Let $\log_2 x + \log_4 x + \log_8 x = \log_k x$ for all $x \in R^+$. If $k = \sqrt[b]{a}$ where $a, b \in N$ then find the smallest positive value of $(a + b)$.

50. Solve the following inequalities:

(*i*) $\log_x(4x - 3) \geq 2$

(*ii*) $\log_{(3x^2+1)} 2 < \frac{1}{2}$

(*iii*) $\log_{x^2}(2 + x) < 1$

51. Find the value of the expression $\frac{2}{\log_4(2000)^6} + \frac{3}{\log_5(2000)^6}$.

52. Solve for x: $\log 4 + \left(1 + \frac{1}{2x}\right)\log 3 = \log\left((3)^{1/x} + 27\right)$

53. Simplify:

(*i*) $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409}\left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6}\right) = ?$

(*ii*) $5^{\log_{1/5}\frac{1}{2}} + \log_{\sqrt{2}}\left(\frac{4}{\sqrt{7}+\sqrt{3}}\right) + \log_{1/2}\left(\frac{1}{(10+2\sqrt{21})}\right) = ?$

(*iii*) $\sqrt{10^{2+\frac{1}{2}\log(16)}} = ?$

54. $\log_{1/5}\frac{4x+6}{x}\geq 0$

55. $\log_7\left(\frac{2x-6}{2x-1}\right)>0$

56. $\log_3|3-4x|>2$

57. If $N=7^{\log_{49}900}$, $A=2^{\log_2 4}+3^{\log_2 4}+4^{\log_2 2}-4\log_2 3$, $D=(\log_5 49)(\log_7 125)$

Find $P=\log_{\left(A-\frac{N}{10}\right)}|N+A+D+6|-\log_5 2$,

58. Solve for x:

(*i*) $\log_2(\log_{1/2}(x))<2$ (*ii*) $\log_{1/2}(\log_3(x))>3$

(*iii*) $(\log_2(x)-1)(\log_3(x)-2)\leq 0$

(*iv*) $(\log_2(x)-1)(\log_{1/2}(x)-2)\leq 0$

59. Solve for x:

(*i*) $\frac{\log_{10}(x-3)}{\log_{10}(x^2-21)}=\frac{1}{2}$ (*ii*) $\log_2(x+1)>\log_3(x+1)$

(*iii*) $\log_{1/2}(x-1)>\log_{1/3}(x-1)$

60. If $a+b+c=1$, $a^2+b^2+c^2=9$, $a^3+b^3+c^3+1$, then find value of $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$.

61. If $x^2+\frac{1}{x^2}=102$, then possible values of $x-\frac{1}{x}$ will be

62. If $x+\frac{1}{x}=\sqrt{5}$ then find

(*i*) $x^2+\frac{1}{x^2}$ (*ii*) $x^3+\frac{1}{x^3}$

(*iii*) $x^4+\frac{1}{x^4}$ (*iv*) $x^6+\frac{1}{x^6}$

63. $a+b+c=10$ and $ab+bc+ac=20$ then find the value of $a^3+b^3+c^3-3abc$

64. If a, b, c are real and $a^2+16b^2+25c^2-4ab-20bc-5ac=0$ then find the ratio of a, b and c.

65. Prove that $(a-b)^3+(b-c)^3+(c-a)^3=3(a-b)(b-c)(c-a)$

66. The value of $\sqrt{5\sqrt{5\sqrt{5\sqrt{5.....}}}}$ - is

67. If $x=\sqrt[3]{7+5\sqrt{2}}-\frac{1}{\sqrt[3]{7+5\sqrt{2}}}$, then the value of $x^3+3x-14$ is equal to

68. $\left[\left\{\left(\frac{1}{7^2}\right)^{-2}\right\}^{-1/3}\right]^{1/4}=7^x$ then x

69. If $\log_{x-3}(2x-3)$ is a meaningful quantity then find the interval in which x must lie.

70. Number of cyphers after decimal before a significant figure starts in $\left(\frac{5}{4}\right)^{-100}$ is equal to [**Use:** $\log_{10}2=0.3010$]

71. Number of real solution of $\log_5[2+\log_3(x+3)]=0$ is

72. Prove that $a^2+b^2+c^2-ab-bc-ac$ is non-negative for all real values of a, b and c. Also prove that equality holds when $a=b=c$.

73. If $4^A+9^B=10^C$, where $A=\log_{16}4$, $B=\log_3 9$ & $C=\log_x 83$, then find x.

74. Prove that $\frac{(a^2-b^2)^3+(b^2-c^2)^3+(c^2-a^2)^3}{(a-b)^3+(b-c)^3+(c-a)^3}=(a+b)(b+c)(c+a)$

75. $a^{\frac{\log_b(\log_b N)}{\log_b a}}$

76. $x>\sqrt{1-x}$

77. $\sqrt{x-2}\geq -1$

78. $(x^2-4)\sqrt{x^2-1}<0$

79. Solve for x:

(*i*) $\frac{1}{2}\leq|2x-1|\leq\frac{3}{5}$ (*ii*) $-\frac{1}{3}\leq|3x-4|\leq 2$

(*iii*) $2\leq|4-5x|\leq\frac{10}{3}$ (*iv*) $3\leq|x^2-1|\leq 8$

80. Solve for x:

(*i*) $|||x+5|-3|-1|=2$

(*ii*) $||||x-5|-7|-3|-2|=1$

81. Solve for x:

(*i*) $2\leq|||x-7|-3|+2|\leq 5$

(*ii*) $||||x-5|-7|-3|-2|\leq 1$

82. Solve for x:

(*i*) $\frac{(|x|-1)}{(|x|-2)}\leq 0$ (*ii*) $\frac{(|x|-1)(|x|-3)}{|x|^2-2|x|}\geq 0$

(*iii*) $\frac{(|x|^2-5|x|+6)}{(4-|x|^2)}\leq 0$

(*iv*) $(||x-1|-2|-3)(|x-2|-3)\geq 0$

(*v*) $(|||x-1|-2|-1|-2)(|x-2|)\geq 0$

83. Solve for x:

(*i*) $\left|\frac{x-3}{x+1}\right|\leq 1$ (*ii*) $\left|\frac{x+1}{x}\right|+|x+1|=\frac{(x+1)^2}{|x|}$

(*iii*) $\left|1+\frac{3}{x}\right|>2$ (*iv*) $|2^x-1|+|4-2^x|<3$

(*v*) $\left(\frac{1}{3}\right)^{\frac{|x+2|}{2-|x|}}>9$

84. Solve: $||x^2-2x+6|-|x+6||=|x^2-3x|$

85. Solve $\frac{5^x(\sin x-\pi)}{(x^2-4)}\geq 0$.

Exercise-3 (JEE Advanced Level)

1. $\log_{10}(\log_2 3)+\log_{10}(\log_3 4)+\log_{10}(\log_4 5)+...+\log_{10}(\log_{1023} 1024)$ simplifies to

(*a*) a composite

(*b*) a prime number

(*c*) rational which is not an integer

(*d*) an integer

2. The expression $\log_p \underbrace{\log_p \sqrt[p]{\sqrt[p]{\sqrt[p]{.....\sqrt[p]{p}}}}}_{n \text{ radical sign}}$, where $p \geq 2$, $p \in N$; $n \in N$ when simplifies is

(*a*) Independent of p

(*b*) Independent of p and of n

(*c*) Dependent on both p and n

(*d*) Positive

3. The solution set of the inequality $\log_{\sin\left(\frac{\pi}{3}\right)}(x^2-3x+2) \geq 2$ is

(*a*) $\left(\frac{1}{2},2\right)$ (*b*) $\left(1,\frac{5}{2}\right)$

(*c*) $\left[\frac{1}{2},1\right)\cup\left(2,\frac{5}{2}\right]$ (*d*) (1, 2)

4. If $\log_{0.5}\log_5(x^2-4) > \log_{0.5} 1$, then '$x$' lies in the interval

(*a*) $(-3,-\sqrt{5})\cup(\sqrt{5},3)$ (*b*) $(-3,-\sqrt{5})\cup(\sqrt{5},2)$

(*c*) $(\sqrt{5},3\sqrt{5})$ (*d*) ϕ

5. The number of positive integers not satisfying the inequality $\log_2(4^x - 2.2^x + 17) > 5$.

(*a*) 2 (*b*) 3

(*c*) 4 (*d*) 1

6. If a, b, c are positive real numbers such that

$a^{\log_3 7}=27; b^{\log_7 11}=49$ and $c^{\log_{11}25}=\sqrt{11}$.

The value of $\left(a^{(\log_3 7)^2}+b^{(\log_7 11)^2}+c^{(\log_{11}25)^2}\right)$ equals

(*a*) 489 (*b*) 469

(*c*) 464 (*d*) 400

7. If $\sqrt{\log_4\{\log_3\{\log_2(x^2-2x+a)\}\}}$ is defined $\forall\, x \in R$, then the set of values of 'a' is

(*a*) [9, ∞) (*b*) [10, ∞)

(*c*) [15, ∞) (*d*) [2, ∞)

8. Find the value of x that satisfies the equation $\log\left(\frac{x^{1/x}}{x^{1/(x+1)}}\right)=\frac{1}{5050}$

(*a*) 1 (*b*) 10 (*c*) 100 (*d*) 1000

9. Number of positive solution which satisfy the equation $\log_2 x \cdot \log_4 x \ \log_6 x = \log_2 x \cdot \log_4 x + \log_2 x \cdot \log_6 x + \log_4 x \cdot \log_6 x$?

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) infinite

10. Sum of all the real solutions of the inequality

$\frac{(x^2+2)(\sqrt{x^2-16})}{(x^4+2)(x^2-9)} \leq 0$ is

(*a*) 5 (*b*) 4 (*c*) 8 (*d*) 0

11. Number of integers, which satisfy the inequality, $\frac{(16)^{1/x}}{(2^{x+3})} > 1$ is equal to:

(*a*) Infinite (*b*) Zero

(*c*) 1 (*d*) 4

12. The set of values of x satisfying simultaneously the inequalities $\frac{\sqrt{(x-8)(2-x)}}{\log_{0.3}\left(\frac{10}{7}(\log_2 5-1)\right)} \geq 0$ and $2^{x-3}-31>0$ is:

(*a*) a unit set

(*b*) an empty set

(*c*) an infinite set

(*d*) a set consisting of exactly two elements.

13. The solution set of inequality $\frac{(3^x-4^x)\cdot\ln(x+2)}{x^2-3x-4} \leq 0$ is

(*a*) $(-\infty, 0]\cup(4,\infty)$ (*b*) $(-2, 0]\cup(4,\infty)$

(*c*) $(-1, 0]\cup(4,\infty)$ (*d*) $(-2,-1]\cup(-1,0)\cup(4,\infty)$

14. Number of values of x in the interval (0, 5) satisfying the equation $\frac{\ln(\sqrt{\sqrt{x^2+1}+x})+\ln(\sqrt{\sqrt{x^2+1}-x})}{\ln x}=x,$ is

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 0

15. The value of x satisfying the equation $(\sqrt{5})^{\log_5 5^{\log_{5^5}5^{\log_5(x)}}}=2$ is

(*a*) 1 (*b*) 2 (*c*) 4 (*d*) 8

16. If $A=\log_{\sqrt{5}}\left(\left(5^{\frac{1}{2}}\right)^{\frac{1}{2}}\right)^{\frac{1}{2}\cdot\cdot^{10\text{ times}}}$, then value of $\log_{\sqrt{3}}(1024A+1)$, is equal to

(*a*) 1 (*b*) 3 (*c*) 5 (*d*) 2

17. If $|x-3|^2-6|x-3|-23<4$, then

(*a*) $x \in (-12, 6)$

(*b*) 17 integers satisfy the inequality

(*c*) 11 non-negative integers satisfy the inequality

(*d*) 6 negative integers satisfy the inequality

18. The roots of the equations $|x| = 49^{\left(\frac{1}{2}+\log_{\frac{1}{7}} 27+\log_{343} 81\right)}$ include

(*a*) One positive number greater than 1 only

(*b*) Two real number

(*c*) Two irrational number

(*d*) One negative rational number

19. If x_1, x_2 are the solution of the inequality $(x-4)^2 + (x-6)^4 = 34$, then

(*a*) $x_1 + x_2 = 10$

(*b*) $x_1 x_2 = 21$

(*c*) $|x_1 - x_2| = 2\sqrt{2}$

(*d*) Both x_1, and x_2 are irrational

20. Which of the following is true?

(*a*) $(\log_{10}2)^2 + 1 > \log_{10}4$

(*b*) $\log_{10}90 > \log_5 50$

(*c*) $\log_4 \log_3 \log_2 16 > \log_{16} 4$

(*d*) $2(\log_{10} 3)^2 - 3(\log_{10} 2)^2 > (\log_{10} 2) \times (\log_{10} 3)$

21. Indicate all correct alternatives, where base of the log is 2.

The equation $x^{(3/4)(\log_2 x)^2 + \log_2 x - (5/4)} = \sqrt{2}$ has:

(*a*) At least one real solution

(*b*) Exactly three real solutions

(*c*) Exactly one irrational solution

(*d*) Imaginary roots

22. The solution set of the system of equations $\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x+y) = \frac{2}{3}$ is

(*a*) (6, 3) (*b*) (3, 6)

(*c*) (6, 12) (*d*) (12, 6)

23. The equation $x^{\left[(\log_3 x)^2 - \frac{9}{2}\log_3 x + 5\right]} = 3\sqrt{3}$ has

(*a*) Exactly three real solution

(*b*) At least one real solution

(*c*) Exactly one irrational solution

(*d*) Complex roots

24. Solution set of the inequality

$$(\log_2 x)^4 - \left(\log_{1/2}\frac{x^3}{8}\right)^2 + 9\log_2\left(\frac{32}{x^2}\right) < 4(\log_{1/2} x)^2 \text{ is}$$

$(a, b) \cup (c, d)$ then the correct statement is

(*a*) $a = 2b$ and $d = 2c$

(*b*) $b = 2a$ and $d = 2c$

(*c*) $\log_c d = \log_b a$

(*d*) there are 4 integers in (c, d)

COMPREHENSION BASED QUESTIONS

Comprehension (Q. No. 25 to 27): Let α and β are the solutions of the equation $\left(\sqrt{x}\right)^{\log_5 x - 1} = 5$ where $\alpha \in I$ and $\beta \in Q$ Then

[**Use:** $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$]

25. The number of significant digits before decimal in $(\alpha)^{10}$ is

(*a*) 13 (*b*) 14

(*c*) 15 (*d*) None of these

26. Number of zeroes after decimal before a significant digit in $(\beta)^{10}$ is

(*a*) 5 (*b*) 7 (*c*) 8 (*d*) 6

27. The value of $(\beta)^{\log_{25} 9}$ is

(*a*) $\frac{1}{3}$ (*b*) 5

(*c*) $\frac{1}{5}$ (*d*) 9

28. Match the Column:

	Column-I		Column-II
A.	The value(s) of x, which does not satisfy the equation $\log_2^2(x^2 - x) - 4\log_2(x-1)\log_2 x = 1$, is (are)	p.	2
B.	The value of x satisfying the equation $2^{\log_2 e^{\ln 5^{\log_5 7^{\log_7 10^{\log_{10}(8x-3)}}}}} = 13$, is	q.	3
C.	The number $N = \left(\frac{1}{\log_2 \pi} + \frac{1}{\log_6 \pi}\right)$ is less than	r.	4
D.	Let $l = (\log_3 4 + \log_2 9)^2 - (\log_3 4 - \log_2 9)^2$ and $m = (0.8)\left(1 + 9^{\log_3 8}\right)^{\log_{65} 5}$ then $(l + m)$ is divisible by	s.	5
		t.	6

(*a*) A → r, t, s; B → q; C → r, s, q; D → q, r

(*b*) A → q, r, s, t; B → p; C → q, r; D → r, s

(*c*) A → q, r, s, t; B → p; C → q, r, s, t; D → q, r, s

(*d*) A → t; B → s; C → q, t; D → r, s

29. Match the columns:

	Column-I		Column-II
A.	If $p = \frac{3\sqrt{2}+2\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}$ then $\log_{(5+2\sqrt{6})} p$ is	p.	0
B.	If $r = \frac{3\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ then $\log_{9+2\sqrt{15}}(1/r)$ is	q.	2
C.	If $t = \frac{3+\sqrt{6}}{5\sqrt{3}-2\sqrt{12}-\sqrt{32}+\sqrt{50}}$ then $\log_{\sqrt{3}} t^2$ is	r.	–1
D.	If $k = \frac{3\sqrt{2}}{\sqrt{3}+\sqrt{6}} - \frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}} + \frac{\sqrt{6}}{\sqrt{2}+\sqrt{3}}$ then $\log_e(k+1)$ is	s.	1

(*a*) A → t, B → s, C → r, D → q

(*b*) A → s, B → r, C → q, D → p

(*c*) A → r, B → p; C → r, D → s

(*d*) A → t, B → q, C → s, D → r

30. Let $f(x)=\dfrac{x^2-6x+5}{x^2-5x+6}$

Column I		Column II	
A.	If $-1<x<1$, then $f(x)$ satisfies	p.	$0<f(x)<1$
B.	If $1<x<2$, then $f(x)$ satisfies	q.	$f(x)<0$
C.	If $3<x<5$, then $f(x)$ satisfies	r.	$f(x)>0$
D.	If $x>5$, then $f(x)$ satisfies	s.	$f(x)<1$

(*a*) A → p, B → q, C → q, D → s
(*b*) A → q, B → s, C → r, D → q
(*c*) A → s, B → r; C → q, D → s
(*d*) A → p, B → q, C → s, D → r

SUBJECTIVE BASED QUESTIONS

31. Find the number of integral solution of the equation $\log_{\sqrt{x}}\left(x+|x-2|\right)=\log_x(5x-6+5|x-2|)$.

32. If *a*, *b* are co-prime numbers and satisfying $\left(2+\sqrt{3}\right)^{\frac{1}{\log_a(2-\sqrt{3})}+\frac{1}{\log_b\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)}}=\dfrac{1}{12}$, then $(a+b)$ can be is equal to

33. If *p* is the smallest value of *x* satisfying the equation $2^x+\dfrac{15}{2^x}=8$ then the value of 4^p is equal to

34. Positive numbers *x*, *y* and *z* satisfy $xyz=10^{81}$ and $(\log_{10}x)(\log_{10}yz)+(\log_{10}y)(\log_{10}z)=468$.
Find the value of
$\left(\log_{10}x\right)^2+\left(\log_{10}y\right)^2+\left(\log_{10}z\right)^2$

35. If (x_1, y_1) and (x_2, y_2) are the solution of the system of equation
$\log_{225}(x)+\log_{64}(y)=4$
$\log_x(225)-\log_y(64)=1$,
then find the value of $\log_{30}(x_1y_1x_2y_2)$.

36. Number of ordered pairs (α, β), $\alpha,\beta \in N$ such that $\alpha^2=4002+\beta^2$ is

37. The sum of all integral values of *x* satisfying the equation $2\log_8(2x)+\log_8(x^2-2x+1)=\dfrac{4}{3}$ is.

38. Suppose n be an integer greater than 1. Let $a_n=\dfrac{1}{\log_n 2002}$.
Suppose $b = a2 + a3 + a4 + a5$ and $c = a10 + a11 + a12 + a13 + a14$. Then find the value of $(b - c)$.

39. If $\log_b a.\ \log_c a+\log_a b.\ \log_c b+\log_a c.\ \log_b c=3$ (where *a*, *b*, *c* are different positive real numbers ≠ 1), then find the value of *a b c*.

40. If the product of alll solutions of the equation $\dfrac{(2009)x}{2010}=(2009)^{\log_x(2010)}$ can be expressed in the lowest form as $\dfrac{m}{n}$ then the value of $(m - n)$ is

41. If the complete solution set of the inequality $(\log_{10}x)^2 \geq \log_{10}x+2$ is $(0, a] \cup [100, \infty)$ then find the value of *a*.

42. The complete solution set of the inequality
$\dfrac{1}{\log_4\dfrac{x+1}{x+2}}<\dfrac{1}{\log_4(x+3)}$, is (–a, ∞), then determine '*a*'.

43. If complete solution set of inequality $\log_{1/2}(x+5)^2>\log_{1/2}(3x-1)^2$ is $(-\infty, p)\cup(q, r)\cup(s, \infty)$ then find $\dfrac{p^2+q^2+r^2}{s^2}$

44. Find the real solutions to the system of equations
$\log_{10}(2000xy)-\log_{10}x\cdot\log_{10}y=4$
$\log_{10}(2yz)-\log_{10}y\cdot\log_{10}z=1$
and $\log_{10}(zx)-\log_{10}z\cdot\log_{10}x=0$

45. Solve the equation $x^{0.5\log_{\sqrt{x}}(x^2-x)}=3^{\log_9 4}$.

46. Solve for *x*: $\log^2(4-x)+\log(4-x)\cdot\log\left(x+\dfrac{1}{2}\right)-2\log 2\log^2\left(x+\dfrac{1}{2}\right)=0$.

47. Solve for *x*
(*a*) $\log_{2x+3}(x^2)<\log_{2x+3}(2x+3)$
(*b*) $\log_{x+3}(x^2-x)<1$

48. Solve for *x* $\log_{0.2}(x^2-x-2)>\log_{0.2}(-x^2+2x+3)$

49. Solve for *x* $(0.3)^{\log_{1/3}\log_2\frac{3x+6}{x^2+2}}>1$

50. Solve for *x* $\log_{0.5}\left(\log_6\dfrac{x^2+x}{x+4}\right)<0$

51. Solve for *x* $\log_3\dfrac{|x^2-4x|+3}{x^2+|x-5|}\geq 0$

52. If *x* and *y* satisfying both the equations
$\log_3 x+\log_2 y=2$;
$3^x-2^y=23$
then sum of all the values of *x* and *y* is

53. For the equation
$(0.4)^{\log^2 x+1}=(6.25)^{2-p\log x}$
(base 10)
If $p = 2$, number of real roots *m*,
If $p = 3$, number of real roots *n*,
Then $m + n =$

54. Solve for *x*:
(*i*) $\log_x(2x-1)>0$
(*ii*) $\log_{(1-x)}(2x-1)<0$
(*iii*) $\log(\log x)+\log(\log x^3-2)=0$

55. Solve for x:

(*i*) $\log_x (2) \cdot \log_{2x} 2 = \log_{4x} 2$

(*ii*) $5^{\log_a x} + 5x^{\log_a 5} = 3 (a > 0)$

(*iii*) $x^{(\log x) + 4} = 32$

(*iv*) $\log_{x+1} (x^2 + x - 6)^2 = 4$

(*v*) $|x-1|^{(\log x)^2 - \log x^2} = (x-1)^3$

(*vi*) $x + \log_{10} (1 + 2^x) = x \cdot \log_{10} 5 + \log_{10} 6$

56. Prove that $\log_a N \cdot \log_b N + \log_b N \cdot \log_c N + \log_c N \cdot \log_a N = \dfrac{\log_a N \cdot \log_b N \cdot \log_c N}{\log_{abc} N}$

57. Find the product of the positive roots of the equation $\sqrt{2005}(x)^{\log_{2005} x} = x^2$.

58. If $a = \log_{12} 18$ & $b = \log_{24} 54$ then find the value of $ab + 5(a - b)$.

59. If $x = 1 + \log_a bc, y = 1 + \log_b ca, z = 1 + \log_c (ab)$ then prove that $xyz = xy + yz + zx$.

60. Find the number of real solution of $(x^2 + x - 2)^3 + (2x^2 - x - 1)^3 = 27(x^2 - 1)^3$

61. Solve: $\log_3 (\sqrt{x} + |\sqrt{x} - 1|) = \log_9 (4\sqrt{x} - 3 + 4|\sqrt{x} - 1|)$

62. Prove that: $2^{\left(\sqrt{\log_a \sqrt[4]{ab} + \log_b \sqrt[4]{ab}} - \sqrt{\log_a \sqrt[4]{\frac{b}{a}} + \log_b \sqrt[4]{\frac{a}{b}}}\right) \cdot \sqrt{\log_a b}}$

$$= \begin{cases} 2 & \text{if } \ b \geq a > 1 \\ 2^{\log_a b} & \text{if } \ 1 < b < a \end{cases}$$

63. Find all the solutions of the equation $|x-1|^{(\log x)^2 - \log x^2} = |x - 1|^3$, where base of logarithm is 10.

64. Solve the following equations for x & y: $\log_{100} |x + y| = \dfrac{1}{2}$

$\log_{10} y - \log_{10} |x| = \log_{100} 4$.

65. Simplify: $\log_{1/3} \sqrt[4]{729 \cdot \sqrt[3]{9^{-1} \cdot 27^{-4/3}}}$

66. Solve for x $-1 \leq \dfrac{1 + x^2}{2x} \leq 1$

67. Solve for x $\dfrac{(x - \sin 1)(x - \sin 2)}{(x - \sin 3)(x - \sin 4)} \leq 0$

68. Solve for x $\dfrac{(|x| - 1)(|x| - 3)}{(|x| - 4)(|x| - 7)} \geq 0$

69. Solve for x $\sqrt{(x-1)}(x-2)x \leq 0$

70. Solve for x $\sqrt{x-5} - \sqrt{9-x} > 1 : x \in Z$

71. Solve for x $\sqrt{x-1} > \sqrt{3-x}$

72. If $\dfrac{(x-3)^{\frac{-|x|}{x}} \sqrt{(x-4)^2}(\pi - x)}{\sqrt{-x}(-x^2 + x - 1)(|x| - 9)} < 0$ then no. of integers x satisfying the inequality is

Exercise-4 (Past Year Questions)

JEE MAIN

1. The sum of the solutions of the equation $\left|\sqrt{x} - 2\right| + \sqrt{x}\left(\sqrt{x} - 4\right) + 2 = 0, (x > 0)$ is equal to: **(2019)**

(*a*) 4 (*b*) 9

(*c*) 10 (*d*) 12

2. The number of real roots of the equation $5 + |2^x - 1| = 2^x (2^x - 2)$ is **(2019)**

(*a*) 3 (*b*) 2

(*c*) 4 (*d*) 1

3. The sum of the solutions of the equation **(2019)** $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0, (x > 0)$ is equal to

(*a*) 9 (*b*) 12

(*c*) 4 (*d*) 10

4. Solution set of $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$ contains **(2020)**

(*a*) Singleton set

(*b*) Two elements

(*c*) At least four elements

(*d*) Infinite elements

5. The product of the roots of the equation **(2020)** $9x^2 - 18|x| + 5 = 0$, is :

(*a*) $\dfrac{25}{9}$ (*b*) $\dfrac{25}{81}$

(*c*) $\dfrac{5}{9}$ (*d*) $\dfrac{5}{27}$

6. The number of solutions of the equation $\log_4(x - 1) = \log_2(x - 3)$ is **(2021)**

7. The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \dfrac{27}{4}$ is **(2021)**

8. The value of $3 + \cfrac{1}{4 + \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{3 + \ldots \infty}}}}$ is equal to **(2021)**

(*a*) $1.5+\sqrt{3}$ (*b*) $2+\sqrt{3}$

(*c*) $3+2\sqrt{3}$ (*d*) $4+\sqrt{3}$

9. The number of elements in the set **(2021)**
$\{x \in \mathbb{R} : (|x| - 3)\,|x + 4| = 6\}$ is equal to

(*a*) 3 (*b*) 2

(*c*) 4 (*d*) 1

10. If α, β are the roots of the equation

$$x^2-\left(5+3^{\sqrt{\log_3 5}}-5^{\sqrt{\log_5 3}}\right)x+3\left(3^{(\log_3 5)^{\frac{1}{3}}}-5^{(\log_5 3)^{\frac{2}{3}}}-1\right)=0$$

then the equation, whose roots are $\alpha+\frac{1}{\beta}$ and $\beta+\frac{1}{\alpha}$, **(2022)**

(*a*) $3x^2 - 20x - 12 = 0$ (*b*) $3x^2 - 10x - 4 = 0$

(*c*) $3x^2 - 10x + 2 = 0$ (*d*) $3x^2 - 20x + 16 = 0$

11. Let $S=\left\{x\in[-6,3]-\{-2,2\}:\frac{|x+3|-1}{|x|-2}\geq 0\right\}$ and $T = \{x \in Z : x^2 - 7\,|x| + 9 \leq 0\}$. Then the number of elements is S $\cap$ T is **(2022)**

(*a*) 7 (*b*) 5

(*c*) 4 (*d*) 3

12. Let p and q be two real numbers such that $p + q = 3$ and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p}+\frac{1}{q}\right)^{-2}$ is equal to _____. **(2022)**

JEE ADVANCED

13. Let (x_0, y_0) be the solution of the following equations $(2x)^{\ln 2} = (3y)^{\ln 3}$

$3^{\ln x} = 2^{\ln y}$

Then x_0 is **(2011)**

(*a*) $\frac{1}{6}$ (*b*) $\frac{1}{3}$

(*c*) $\frac{1}{2}$ (*d*) 6

14. The value of

$$6+\log_{\frac{3}{2}}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}}....}}\right)$$ is

(2012)

15. If $3^x = 4^{x-1}$, then $x =$ **(2013)**

(*a*) $\frac{2\log_3 2}{2\log_3 2-1}$ (*b*) $\frac{2}{2-\log_2 3}$

(*c*) $\frac{1}{1-\log_4 3}$ (*d*) $\frac{2\log_2 3}{2\log_2 3-1}$

16. The value of $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}}\times(\sqrt{7})^{\frac{1}{\log_4 7}}$ is _____ **(2018)**

ANSWER KEY

CONCEPT APPLICATION

1. (*a*) **2.** (*b*) **3.** (*d*) **4.** (*c*) **5.** (*d*) **6.** (*i*) 14 (*ii*) 52 (*iii*) 194 **9.** [0] **10.** [–224]

11. $x^6 - y^6$ **12.** $p = 3/2, q = 1, r = 4/3$ **13.** (*d*) **14.** (*c*) **15.** (*b*) **16.** [60] **17.** [99] **18.** [4]

19. (*d*) **20.** [*2ab*] **21.** $x = \pm 4\sqrt{\frac{5}{4}}$ **22.** $x = \pm\sqrt{\frac{3}{5}}$ **23.** (*c*) **24.** $x \in [-\infty, -3] \cup [-2, 0] \cup [6, 0]$

25. $x \in (-6, 0] \cup [2, 3] \cup (6, \infty) \cup \{4\}$ **26.** [6] **27.** (*d*) **28.** $\left(\frac{1}{5}, \frac{2}{5}\right)$ **29.** (243) **30.** [–5] **31.** [18]

32. [–1] **33.** (*i*) $(1, \infty)$ (*ii*) $[1, \infty)$ (*iii*) (0, 1) (*iv*) (0, 1] (*v*) (0, 1) (*vi*) (0, 1] (*vii*) $(1, \infty)$ (*viii*) $[1, \infty)$ (*ix*) $(3, \infty)$ (*x*) $[3/2, \infty)$

34. (*i*) $x \in \left(\frac{3}{2}, \frac{19}{2}\right)$ (*ii*) $x \in \left(\frac{2}{3}, \frac{17}{24}\right]$ (*iii*) $x \in (4^{16}, \infty)$ (*iv*) $x \in \left(0, \frac{1}{2}\right)$ **35.** {3, –3} **36.** {–2, 2} **38.** $\frac{12 - 4a}{3 + a}$

39. [44] **40.** (*d*) **41.** (*b*, *c*) **42.** (*i*) {–1, 5} (*ii*) {6, –6, 7, –1} (*iii*) (14, –4, 0, 10, 2, 8) **43.** (–6, 8)

44. (*d*) **45.** $(0, 2) \cup (4, \infty)$ **46.** $x \geq 1$ **47.** (*i*) $x \in (-\infty, 1) \cup (1, \infty)$ (*ii*) $x \in (-\infty, -5] \cup [5, \infty)$ (*iii*) $x \in (-7, 7)$ (*iv*) $x \in [-10, 10]$ (*v*) $x \in R$ (*vi*) $x \in \phi$ (*vii*) $x \in R$ (*viii*) $x \in R$ (*ix*) $x \in \phi$

48. (*i*) $x \in (-\infty, 0) \cup (2, \infty)$ (*ii*) $1 < x < 3$ (*iii*) $x \in (-2, -1) \cup (0, 1)$ (*iv*) $x \in (-\infty, 0) \cup (1, \infty)$ (*v*) $-4/3 \leq x \leq 2$ (*vi*) $x \in \phi$

49. (*i*) $-1 \leq x \leq 5$ (*ii*) $x \in (-\infty, -3] \cup [9, \infty) \cup [-1, 7]$ (*iii*) $x \in [-4, 0] \cup [2, 8] \cup [10, 14]$

EXERCISE-1 (TOPICWISE)

1. (*b*)	**2.** (*b*)	**3.** (*d*)	**4.** (*a*)	**5.** (*c*)	**6.** (*a*)	**7.** (*d*)	**8.** (*b*)	**9.** (*b*)	**10.** (*a*)
11. (*b*)	**12.** (*d*)	**13.** (*d*)	**14.** (*c*)	**15.** (*d*)	**16.** (*c*)	**17.** (*b*)	**18.** (*d*)	**19.** (*c*)	**20.** (*c*)
21. (*d*)	**22.** (*b*)	**23.** (*c*)	**24.** (*c*)	**25.** (*d*)	**26.** (*b*)	**27.** (*a*)	**28.** (*b*)	**29.** (*c*)	**30.** (*a*)
31. (*b*)	**32.** (*d*)	**33.** (*b*)	**34.** (*a*)	**35.** (*d*)	**36.** (*c*)	**37.** (*b*)	**38.** (*d*)	**39.** (*b*)	**40.** (*d*)

EXERCISE-2 (LEARNING PLUS)

1. (*c*)	**2.** (*a*)	**3.** (*a*)	**4.** (*a*)	**5.** (*b*)	**6.** (*c*)	**7.** (*a*)	**8.** (*b*)	**9.** (*d*)	**10.** (*c*)
11. (*d*)	**12.** (*b*)	**13.** (*c*)	**14.** (*a*)	**15.** (*b*)	**16.** (*d*)	**17.** (*b*)	**18.** (*a*)	**19.** (*a*)	**20.** (*d*)
21. (*b*)	**22.** (*b*)	**23.** (*c*)	**24.** (*d*)	**25.** (*c*)	**26.** (*b*)	**27.** (*c*)	**28.** (*d*)	**29.** (*a*)	**30.** (*c*)
31. (*c*)	**32.** (*a*)	**33.** (*a*)	**34.** (*d*)	**35.** (*b*)	**36.** (*d*)	**37.** (*a*)	**38.** (*a*)	**39.** (*b*)	**40.** (*d*)

41. (*b*) **42.** (*d*) **43.** (*d*) **44.** (*a*) **45.** (*b*) **46.** (*c*) **47.** (e^{-3})

48. $x = 64, y = 16, z = 9$ **49.** 75 **50.** (*i*) $x \geq 3$, (*ii*) $|x| > 1$ (*iii*) $x \in (-2, 1)\ u\ (2, \infty)$ **51.** $\left[\frac{1}{6}\right]$ **52.** (ϕ)

53. (*i*) $\to 1$ (*ii*) $\to 6$ (*iii*) $\to 20$ **54.** $\left[-2, \frac{3}{2}\right)$ **55.** $\left(-\infty, \frac{1}{2}\right)$ **56.** $\left(-\infty, \frac{1}{2}\right)$ **57.** [2]

58. (*i*) $x \in \left(1, 3^{\frac{1}{8}}\right)$, (*ii*) $x \in (1, 3^{1/8})$, (*iii*) $x \in (2, 9)$, (*iv*) $x \in \left(0, \frac{1}{4}\right] \cup [2, \infty)$

59. (*i*) $x \in \{5\}$, (*ii*) $x \in (0, \infty)$, (*iii*) $x \in (1, 2)$ **60.** [1] **61.** [±10] **62.** (*i*) [3], (*ii*) [$2\sqrt{5}$], (*iii*) [7], (*iv*) [18]

63. [4*w*] **64.** (20 : 5 : 4) **66.** [5] **67.** [0] **68.** $\left[-\frac{1}{3}\right]$ **69.** $x \in (3, 4) \cup (4, \infty)$ **70.** [9] **71.** [4]

73. [10] **75.** $\log_b N$ **76.** $x \in \left(\frac{\sqrt{5} - 1}{2}, 1\right)$ **77.** $x \in (2, \infty)$ **78.** $x \in (-2, -1) \cup (1, 2)$

79. (*i*) $x \in \left(\frac{1}{5},\frac{1}{4}\right) \cup \left[\frac{3}{4},\frac{4}{5}\right]$ (*ii*) $x \in \left[\frac{2}{3},2\right]$ (*iii*) $x \in \left[\frac{2}{15},\frac{2}{5}\right] \cup \left[\frac{6}{5},\frac{22}{15}\right]$ (*iv*) $x \in (-3, -2) \cup (2, 3)$

80. (*i*) $x \in \{-11, -5, -1\}$ (*ii*) $x \in \{-8, -6, -2, -4, 0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$

81. (*i*) $x \in [1, 13]$ (*ii*) $x \in [-8, -6] \cup [-4, -2] \cup [0, 2] \cup [4, 6] \cup [8, 10] \cup [12, 14] \cup [16, 18]$

82. (*i*) $x \in (-2, -1] \cup [1, 2)$ (*ii*) $x \in (-\infty, -3] \cup (-2, -1] \cup [3, \infty) \cup [1. 2)$ (*iii*) $x \in (-\infty, -3] \cup [3, \infty)$
(*iv*) $x \in (-\infty, -4] \cup [-1, 5] \cup [6, \infty)$ (*v*) $x \in (-\infty, -4] \cup [6, \infty] \cup [6, \infty) \cup \{2\}$

83. (*i*) $x \in [1, \infty]$ (*ii*) $x \in [0, \infty] \cup \{-1\}$ (*iii*) $x \in (-1, 0) \cup (0, 3)$ (*iv*) $x \in \phi$ (*v*) $x \in (2, 6)$

84. $[-6, \infty)$ **85.** $x \in (-2, 2)$

EXERCISE-3 (JEE ADVANCED LEVEL)

1. (*d*) **2.** (*a*) **3.** (*c*) **4.** (*a*) **5.** (*a*) **6.** (*b*) **7.** (*b*) **8.** (*c*) **9.** (*c*) **10.** (*d*)
11. (*a*) **12.** (*a*) **13.** (*d*) **14.** (*a*) **15.** (*c*) **16.** (*d*) **17.** (*b, c*) **18.** (*b, c*) **19.** (*a, c, d*)
20. (*a, d*) **21.** (*a, b, c*) **22.** (*a, b*) **23.** (*a, b, c, d*) **24.** (*b, c*) **25.** (*b*) **26.** (*d*) **27.** (*a*) **28.** (*c*)
29. (*b*) **30.** (*a*) **31.** [1] **32.** [2] **33.** [7] **34.** [3] **35.** [9] **36.** [5625] **37.** [12] **38.** [–1]
39. [1] **40.** [1] **41.** [0.1] **42.** [1] **43.** [17/3] **44.** [$x = 1; y = 5; z = 1$; or $x = 100; y = 20; z = 100$]
45. [2] **46.** [7/4] **47.** (*a*) $\left[x \in \left(\frac{-3}{2},3\right) - \{-1,0\}\right]$ (*b*) $[x \in (-3, -2) \cup (-1, 0) \cup (1, 3)]$ **48.** $[x \in (2, 5/2)]$
49. (–1/2, 2) **50.** $[(-4, -3) \cup (8, \infty)]$ **51.** $(-\infty, 2/5]$ **52.** [5]
54. (*i*) $[(1/2, \infty) - \{1\}]$ (*ii*) $[(1/2, 1)]$ (*iii*) [10] **55.** (*i*) [2] (*ii*) [3/2] (*iii*) [2.22] (*iv*) [2.022] (*v*) [1]
57. $[(2008)^2]$ **58.** [1] **61.** $[x \in [0, 1] \cup \{4\}]$ **63.** [1000 or 1/10] **64.** $\left[x \in \left(\frac{10}{3},\frac{20}{3}\right);\ y \in (-10, 20)\right]$
65. [–1] **66.** [1, –1] **67.** $[x \in (\sin 1, \sin 3)]$ **68.** $x \in (-\infty, -7] \cup (-4, -3] \cup [-1, 1] \cup [3, 4] \cup [7, \infty]$
69. $x \in (1, 2)$ **70.** $x \in (8.323,9)$ **71.** $x \in (2, 3)$ **72.** [31]

EXERCISE-4 (PAST YEAR QUESTIONS)

JEE Main

1. (*c*) **2.** (*d*) **3.** (*d*) **4.** (*a*) **5.** (*b*) **6.** [1] **7.** [2] **8.** (*a*) **9.** (*b*) **10.** (*b*)
11. (*d*) **12.** (4)

JEE Advanced

13. (*a*) **14.** (*d*) **15.** (*a, b, c*) **16.** [8]